Lesson 1: Fractions Refresher: Mixed Numbers

Mixed numbers and improper fractions, adding, subtracting and multiplying them, and checking with an estimate and the calculator.

1. Mixed Numbers and Improper Fractions

Measurements in inches, feet, yards and cups often use fractions bigger than 1, like $2\frac{3}{4}$ in. There are two ways to write them.

- A **proper fraction** is less than 1. Its numerator is smaller than its denominator: $\frac{3}{4}$, $\frac{5}{8}$.
- An **improper fraction** is 1 or more. Its numerator is the same as or bigger than its denominator: $\frac{11}{4}$, $\frac{8}{8}$.
- A **mixed number** is a whole number and a proper fraction together: $2\frac{3}{4}$ means $2 + \frac{3}{4}$.

The bars show the same amount both ways. Each whole bar is 4 quarters, so 2 wholes and 3 more quarters make 11 quarters: $2\frac{3}{4} = \frac{11}{4}$.



Improper fraction to mixed number: divide the numerator by the denominator. The answer is the whole number. The remainder goes over the same denominator.

$\frac{17}{5}$: $17 \div 5 = 3$ remainder 2, so $\frac{17}{5} = 3\frac{2}{5}$

Mixed number to improper fraction: multiply the whole number by the denominator, add the numerator, and keep the denominator.

$4\frac{2}{3} = \frac{4\times3+2}{3} = \frac{14}{3}$

A mixed number is in **simplest form** when its fraction part is in simplest form (Lesson 1.2): $3\frac{6}{8} = 3\frac{3}{4}$.

Watch out. When you simplify a mixed number, only the fraction part changes. The whole number stays the same: $3\frac{6}{8} = 3\frac{3}{4}$, not $\frac{3}{4}$ and not $1\frac{1}{2}$.

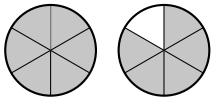
Calculator. Enter $2\frac{3}{4}$ as 2 [A b/c] 3 [A b/c] 4 [ENTER]. The screen shows `2u3⌟4` (the `u` sits between the whole number and the fraction). Press [2nd] [d/e] (the second job of the [A b/c] key) to switch to `11⌟4`. Press it again to switch back. The calculator also puts fractions in simplest form for you.

Example 1

Write each picture as a mixed number and as an improper fraction.

(a) 

Mixed: _____ Improper: _____

(b) 

Mixed: _____ Improper: _____

Example 2

Write each improper fraction as a mixed number in simplest form.

(a) $\frac{23}{4}$ (b) $\frac{45}{8}$ (c) $\frac{38}{6}$ (d) $\frac{30}{12}$

Write each mixed number as an improper fraction.

(e) $3\frac{5}{8}$ (f) $6\frac{2}{3}$ (g) $4\frac{7}{10}$ (h) $12\frac{1}{2}$

Example 3

A breakfast café measures dry ingredients with scoops.

(a) A batch of granola needs $3\frac{1}{2}$ cups of oats. The cook uses a $\frac{1}{4}$-cup scoop. How many scoops is that?

(b) The cook adds 11 scoops of sugar with a $\frac{1}{3}$-cup scoop. How many cups of sugar is that, as a mixed number?

2. Adding Mixed Numbers

You can only add pieces that are the **same size**. So fractions must have the same denominator before you add them. Then add the numerators and keep the denominator: $\frac{3}{8} + \frac{1}{8} = \frac{4}{8} = \frac{1}{2}$.

If the denominators are different, find a **common denominator**: a number that both denominators divide into evenly. The smallest one, the **lowest common denominator**, keeps the numbers small.

Quick refresher. To find a common denominator, count by the bigger denominator until the smaller one divides in. For 4 and 6: 6, 12. 12 works, so use twelfths. Then make equivalent fractions (Lesson 1.2): $\frac{3}{4} = \frac{9}{12}$ and $\frac{1}{6} = \frac{2}{12}$.

To add mixed numbers

Step 1: Give the fraction parts a common denominator.
Step 2: Add the whole numbers. Add the fractions.
Step 3: If the fraction part is improper, **regroup**: change it to a mixed number and add its whole to the whole number.
$5\frac{11}{8} = 5 + 1\frac{3}{8} = 6\frac{3}{8}$
Step 4: Write the answer in simplest form.

Watch out. Never add the denominators. $\frac{1}{4} + \frac{1}{4} = \frac{2}{4}$ (two quarters), not $\frac{2}{8}$.

Example 4

Add. Write each answer in simplest form.

(a) $\frac{3}{8}+\frac{1}{8}$

(b) $\frac{2}{3}+\frac{1}{4}$

(c) $2\frac{1}{6}+3\frac{1}{2}$

(d) $4\frac{3}{4}+1\frac{5}{8}$

Example 5

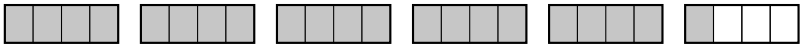
A plumber joins three pieces of copper pipe end to end. They are $14\frac{3}{8}$ in, $9\frac{3}{4}$ in and $6\frac{1}{2}$ in long.

(a) Estimate the total length. Round each piece to the nearest inch.

(b) Find the exact total length. Is it close to your estimate?

3. Subtracting Mixed Numbers

Subtract the same way: common denominator first, then subtract the whole numbers and the fractions. Sometimes the top fraction is too small to take the other fraction away, as in $5\frac{1}{4}-2\frac{3}{4}$. Then **regroup** (borrow): take 1 whole from the whole number and change it into fraction pieces.



$5\frac{1}{4}=4+\frac{4}{4}+\frac{1}{4}=4\frac{5}{4}$. Now take away $2\frac{3}{4}$: $4\frac{5}{4}-2\frac{3}{4}=2\frac{2}{4}=2\frac{1}{2}$.

To subtract mixed numbers

Step 1: Give the fraction parts a common denominator.
Step 2: If the first fraction is smaller than the one you subtract, regroup: take 1 from the whole number and add it to the fraction as $\frac{8}{8}$, $\frac{4}{4}$, $\frac{12}{12}$, ...
Step 3: Subtract the whole numbers. Subtract the fractions.
Step 4: Write the answer in simplest form.

Watch out. Do not flip the fractions to make the subtraction easy. In $5\frac{1}{4}-2\frac{3}{4}$, doing $\frac{3}{4}-\frac{1}{4}$ gives the wrong answer $3\frac{2}{4}$. Regroup instead.
To subtract from a whole number, regroup too: $9=8\frac{16}{16}$.

Example 6

Subtract. Write each answer in simplest form.

(a) $6\frac{7}{8} - 2\frac{3}{8}$

(b) $5\frac{1}{3} - 1\frac{5}{6}$

(c) $9 - 3\frac{5}{16}$

(d) $7\frac{1}{4} - 3\frac{2}{3}$

Example 7

Hailey has $5\frac{1}{4}$ yd of fabric to make costumes for the school play. She uses $1\frac{7}{8}$ yd for a cape and $2\frac{1}{2}$ yd for a curtain.

(a) How much fabric is left?

(b) Is there enough left for a sash that needs $\frac{3}{4}$ yd? Explain.

4. Multiplying with Mixed Numbers

To multiply two fractions, multiply the numerators and multiply the denominators: $\frac{2}{3} \times \frac{4}{5} = \frac{8}{15}$. Remember that "of" means multiply: $\frac{1}{2}$ of $3\frac{1}{3}$ is $\frac{1}{2} \times 3\frac{1}{3}$.

Mixed numbers must be changed to improper fractions first. A whole number is a fraction over 1: $4 = \frac{4}{1}$.

To multiply with mixed numbers

Step 1: Change each mixed number (and whole number) to an improper fraction.
Step 2: Multiply the numerators. Multiply the denominators.
Step 3: Change the answer back to a mixed number in simplest form.

$4 \times 2\frac{3}{8} = \frac{4}{1} \times \frac{19}{8} = \frac{76}{8} = 9\frac{4}{8} = 9\frac{1}{2}$

Watch out. Do not just multiply the whole numbers and the fractions separately. $2\frac{1}{2} \times 2\frac{1}{2}$ is *not* $4\frac{1}{4}$. Change to improper fractions: $\frac{5}{2} \times \frac{5}{2} = \frac{25}{4} = 6\frac{1}{4}$.

Shortcut. You may simplify before you multiply: divide a numerator and a denominator by the same number. $\frac{7}{4} \times \frac{12}{5}$: $12 \div 4 = 3$ and $4 \div 4 = 1$, so it becomes $\frac{7}{1} \times \frac{3}{5} = \frac{21}{5}$. Smaller numbers, same answer.

Example 8

Multiply. Write each answer in simplest form.

(a) $\frac{2}{3} \times \frac{3}{5}$

(b) $4 \times 2\frac{3}{8}$

(c) $\frac{1}{2}$ of $3\frac{1}{3}$

(d) $1\frac{3}{4} \times 2\frac{2}{5}$

Example 9

Jamal's pancake recipe makes 12 pancakes. He wants to make 3 batches for a team breakfast, and half a batch at home. Complete the table.

Ingredient	**1 batch**	**3 batches**	**$\frac{1}{2}$ batch**
Flour	$1\frac{3}{4}$ cups		
Milk	$1\frac{1}{3}$ cups		
Sugar	$2\frac{1}{2}$ tbsp		

5. Estimating and Checking with the Calculator

Before you trust an answer, **estimate**. Round each mixed number to the nearest whole number: if the fraction part is $\frac{1}{2}$ or more, round up; if it is less than $\frac{1}{2}$, round down. Then do the easy calculation in your head.

The calculator can do the exact work. It can also change a mixed number to a decimal, and a decimal back to a fraction. Tape measures and plans use fractions of an inch, so this helps when a calculation gives a decimal.

Estimate (round to whole numbers) → **calculate** (by hand or calculator) → **check** (is the answer close to the estimate?)

A mixed number as a decimal: divide the fraction part, then add the whole: $3\frac{3}{4} = 3 + 0.75 = 3.75$

Calculator. $7\frac{7}{8} - 2\frac{1}{3}$: 7 [A b/c] 7 [A b/c] 8 [−] 2 [A b/c] 1 [A b/c] 3 [ENTER]. The answer shows as a mixed number (the `u` separates the whole number). Press [2nd] [d/e] to see it as an improper fraction.

Fraction ↔ decimal: press [2nd] [F<>D]. A decimal such as `4.625` [ENTER] then [2nd] [F<>D] shows the fraction ($4\frac{5}{8}$).

Example 10

For each one, estimate first. Then find the exact answer as a mixed number and as a decimal (to the nearest hundredth). Check with the calculator.

(a) $7\frac{7}{8} - 2\frac{1}{3}$

(b) $3\frac{2}{5} \times 4\frac{1}{2}$

Example 11

Jesse is building a bench seat from 3 boards laid end to end. Each board is $23\frac{5}{8}$ in long. He says the seat will be $59\frac{7}{8}$ in long.

(a) Use an estimate to show that Jesse's answer is not reasonable.

(b) Find the correct length.

(c) A drawing for the bench says the legs are 16.375 in long. Write this as a mixed number of inches, so Jesse can find it on his tape measure.

Lesson 2: The SI System: Units, Prefixes and Referents

The metric units for length, mass and capacity, what the prefixes mean, everyday objects that help you picture a unit, measuring to the millimetre, and changing units by moving the decimal point.

1. SI Units: What We Measure and With What

The **SI** (short for *Système International*, French for "International System") is the **metric system**. It is the official measurement system in Canada and in most of the world. Each kind of measurement has one **base unit**. Bigger and smaller units are made from it by multiplying or dividing by 10, 100, 1000, ..., just like place value.

What you measure	It tells you . . .	SI unit (symbol)	Other units you will see
Length	how long, wide, tall or far	metre (m)	millimetre (mm), centimetre (cm), kilometre (km)
Mass	how heavy something is	gram (g)	milligram (mg), kilogram (kg), tonne (t)
Capacity	how much a container holds	litre (L)	millilitre (mL)
Temperature	how hot or cold	degree Celsius (°C)	
Area	how much flat surface is covered	square metre (m^2)	square centimetre (cm^2)
Volume	how much space an object takes up	cubic metre (m^3)	cubic centimetre (cm^3)

How the SI units fit together

- Water freezes at 0 °C and boils at 100 °C.
- 1 tonne (t) = 1000 kg.
- A cube 1 cm on each side has a volume of 1 cm^3. It holds exactly **1 mL**, and 1 mL of water has a mass of about **1 g**.
- So **1 L of water has a mass of about 1 kg**. (Area and volume units come later in this unit and in Unit 4.)

Example 1

For each item, write what is being measured (length, mass, capacity, temperature or area) and the SI unit you would use.

Item	What is measured?	SI unit (symbol)
(a) the distance from Calgary to Banff		
(b) how much water a kettle holds		
(c) how heavy a bag of potatoes is		
(d) how warm it is outside in July		
(e) the carpet needed to cover a floor		
(f) the width of a phone screen		

2. The Prefixes

A **prefix** is a word part at the front of a unit. It tells you **how many base units** the unit is worth. The same prefixes work for metres, grams and litres.

Prefix	kilo	hecto	deca	(base unit)	deci	centi	milli
Symbol	k	h	da	m, g, L	d	c	m
Worth	1000	100	10	1	0.1	0.01	0.001
Example	1 km = 1000 m	1 hm = 100 m	1 dam = 10 m	1 m	1 dm = 0.1 m	1 cm = 0.01 m	1 mm = 0.001 m

Read the bottom row the other way, too: 1 m = 10 dm = 100 cm = 1000 mm. A memory trick for the order: **K**ing **H**enry **D**ied **B**y **D**rinking **C**hocolate **M**ilk (kilo, hecto, deca, **B**ase, deci, centi, milli).

The units you will use most

1 km = 1000 m 1 m = 100 cm = 1000 mm 1 cm = 10 mm
1 kg = 1000 g 1 g = 1000 mg 1 L = 1000 mL 1 t = 1000 kg

Watch out. The letter m does two jobs. At the **end** of a symbol it means **metre**; at the **front** of a two-letter symbol it means **milli**. So mm is a millimetre, mg a milligram, mL a millilitre. Write litre with a capital L, write km with a small k, never add an "s" to a symbol, and leave a space: 5 km, not 5km or 5 kms.

Example 2

(a) Complete the table.

Unit name	Symbol	1 of these equals ...
kilometre	km	1000 m
millilitre		
	cm	
kilogram		
	mg	
hectolitre		
	dm	

(b) Fill in the blanks. 1 m = _____ cm = _____ mm 1 cm = _____ mm

1 L = _____ mL 1 kg = _____ g

3. Referents and Choosing a Sensible Unit

A **referent** is a familiar object that is about the size of a unit. A referent helps you **picture** a unit and **estimate** a measurement without a ruler or a scale.

Length		Mass, capacity and temperature	
1 mm	the thickness of a dime	1 g	a paper clip
1 cm	the width of your smallest fingernail	1 kg	a 1 L carton of water or milk
1 m	a long step, or a doorknob's height from the floor	1 t	a small car
2 m	the height of a door	1 mL	about 20 drops of water
1 km	about a 12-minute walk	5 mL	a teaspoon 250 mL: a cup
		1 L	a carton of milk
		°C	0: water freezes 20: room 37: body 100: water boils

To choose a **sensible unit**, pick the unit that gives an easy number, not a huge one or a tiny one. The height of a door is about 2 m. Saying 2000 mm or 0.002 km is correct but not sensible.

- **Length:** mm for very small things (a screw, the thickness of glass); cm for things you hold (a book, a phone); m for rooms, buildings and fields; km for distances between places.
- **Mass:** mg for medicine and vitamins; g for small things and food packages; kg for people, pets and big bags; t for trucks and heavy loads.
- **Capacity:** mL for small containers and spoonfuls; L for jugs, tanks and pools.

Example 3

(a) Choose the most sensible unit: mm, cm, m, km, g, kg, mL or L.

the length of a hockey rink		the mass of a cat	
the thickness of a phone case		the juice in a juice box	
the drive from Calgary to Red Deer		the mass of a slice of cheese	
the length of a pencil		the gas in a car's tank	

(b) Circle the most reasonable measurement.

(i) the height of a basketball hoop: 3 cm 3 m 3 km

(ii) the mass of a large egg: 60 mg 60 g 60 kg

(iii) a bathtub holds: 150 mL 150 L 15 000 L

(iv) the water in a hot bath: 4 °C 40 °C 400 °C

Example 4

Use a referent to estimate.

(a) Tae's smallest fingernail is about 1 cm wide. A pencil is about 17 fingernails long. About how long is the pencil?

(b) A door is about 2 m tall. The gym wall is about 4 doors tall. About how tall is the wall?

(c) A paper clip has a mass of about 1 g. About what is the mass of 250 paper clips?

(d) A milk carton holds 1 L. A soup pot holds about as much as 6 cartons. About how much does the pot hold?

4. Measuring With a Metric Ruler or Tape

On a metric ruler the numbers are **centimetres**. Each centimetre is split into 10 small spaces. Each small space is **1 millimetre**. The middle mark, a bit longer, is at 5 mm (half a centimetre).

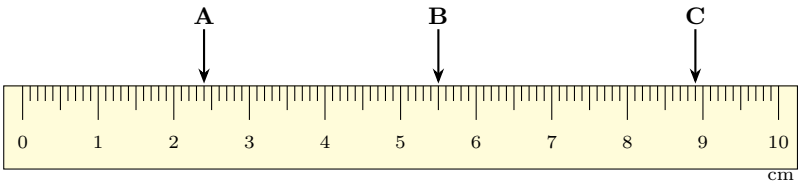
Step 1. Line up one end of the object with the **0 mark** (not the end of the ruler).
Step 2. Read the last full centimetre before the other end.
Step 3. Count the millimetre marks past it. 6 cm and 4 mm is **6.4 cm** or **64 mm**.

If the object does not start at 0, subtract: **length = end reading − start reading**.

A metric **tape measure** works the same way but is longer. Most tapes number every centimetre, so the mark 126 means 126 cm, which is 1 m 26 cm or 1.26 m.

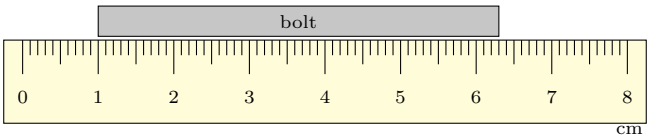
Example 5

(a) Read the ruler at A, B and C. Write each reading in centimetres and in millimetres.

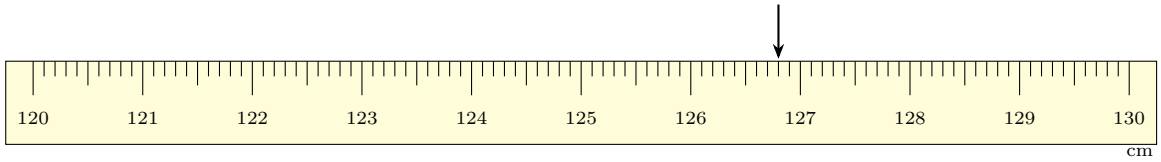


	cm	mm
A		
B		
C		

(b) How long is the bolt? Give your answer in cm and in mm.

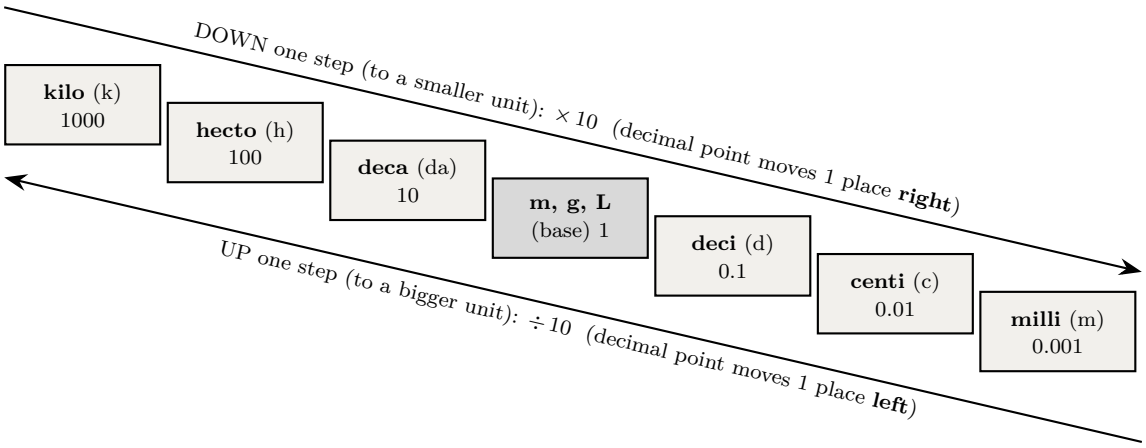


(c) Aiden measures the width of a window with a tape measure. The end of the window is at the arrow. Write the width in cm, in mm and in m.



5. Changing Units: Moving the Decimal Point

SI units are based on 10, so to change units you multiply or divide by 10, 100 or 1000. That just **moves the decimal point**. Use the staircase.



Step 1. Find the unit you have and the unit you want on the staircase.
Step 2. Count the steps between them, and see if you go **down** or **up**.
Step 3. Down: multiply (move the decimal point right one place per step). Up: divide (move it left). Fill empty places with zeros.

For example, metres to centimetres is 2 steps down, so $\times 100$: $3.45 \text{ m} \to 34.5 \to 345 \text{ cm}$.

Smaller unit ⇒ bigger number. **Bigger unit ⇒ smaller number.**
Check every answer: 3.45 m = 345 cm makes sense, because it takes many small centimetres to make a few metres.

Watch out. Count every step, even the ones you do not use much. From m to mm is **3** steps (m, dm, cm, mm), so $\times 1000$, not $\times 100$.

Calculator. 3.45 [×] 100 [ENTER] shows 345. 850 [÷] 1000 [ENTER] shows 0.85.

Example 6

Change each measurement.

Change	**Up or down? How many steps?**	**Work**	**Answer**
(a) 3.45 m to cm			
(b) 2.6 km to m			
(c) 85 mm to cm			
(d) 640 cm to m			
(e) 1250 g to kg			
(f) 0.75 L to mL			
(g) 4 mm to m			
(h) 2.5 cm to mm			

To **compare** measurements in different units, change them to the **same unit** first.

Example 7

(a) Which is longer, 0.8 m or 75 cm?

(b) Which holds more: one 1.5 L bottle of pop, or four 355 mL cans?

(c) Four friends did the long jump. Sana jumped 4.35 m, Theo 428 cm, Uma 4410 mm and Vic 0.0043 km. Put the jumps in order from shortest to longest. Who won?

Example 8

(a) Marisol is selling lemonade at a fundraiser. Each cup holds 250 mL. A jug holds 4 L.
(i) How many cups can she fill from one jug?
(ii) How many jugs does she need to make to fill 50 cups?

(b) Owen runs 3 laps of a 400 m track. Then he runs 1.8 km on the road. How far does he run in all, in kilometres?

Lesson 3: The Imperial System and Reading a Tape Measure

Inches, feet, yards and miles, reading a ruler or tape measure to the nearest sixteenth of an inch, and writing lengths in feet and inches.

1. Imperial Units of Length

Canada's official system of measurement is the **SI** (metric) system. The **imperial system** is an older system. The United States still uses it, and so do many Canadian trades. The imperial units of length are the **inch**, the **foot**, the **yard** and the **mile**.

Unit	Symbol	How it relates	Referent (about this long)
inch	in or ″	the smallest unit	the width of your thumb at the top knuckle
foot	ft or ′	1 ft = 12 in	a school ruler; a sheet of paper is 11 in long
yard	yd	1 yd = 3 ft = 36 in	one long step; a little shorter than a metre stick
mile	mi	1 mi = 1760 yd = 5280 ft	about a 20-minute walk

Watch out. The two marks ″ mean **inches** and one mark ′ means **feet**: 6′ is 6 feet (a tall person), 6″ is 6 inches (half a ruler).

Where Canadians still use imperial units: building (a "2 by 4", sheets of plywood and drywall are 4 ft by 8 ft, wall studs 16 in apart), people's heights, TV and phone screens (measured corner to corner in inches), tire and wheel sizes, plumbing pipe, sports (a CFL field is 110 yd long, golf in yards), and road signs in miles when you drive in the United States.

Example 1

(a) Choose the best imperial unit (inch, foot, yard or mile) to measure each length.

the length of a phone		the length of a football field	
the height of a basement ceiling		the distance from Seattle to Portland	
the width of a wrench		the length of a garden hose	

(b) Circle the most reasonable length.

(i) the height of a classroom door	7 in	7 ft	7 yd	7 mi
(ii) the length of a new pencil	7 in	7 ft	7 yd	7 mi
(iii) the length of a hockey rink	200 in	200 ft	200 yd	200 mi

(c) Use the relationships in the box. Which is longer?

(i) 1 yd or 2 ft? _____ (ii) 1 ft or 10 in? _____ (iii) 1 mi or 2000 yd? _____

2. How an Inch Is Split Up

An imperial ruler splits each inch into 2 halves, then 4 quarters, then 8 eighths, then 16 **sixteenths**. The lines have different lengths so you can tell them apart:

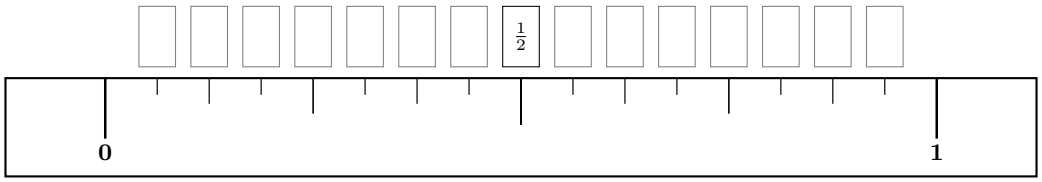
longest line	a whole inch (1, 2, 3, ...)
next longest	a half: $\frac{1}{2}$
next	quarters: $\frac{1}{4}$, $\frac{3}{4}$
next	eighths: $\frac{1}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, $\frac{7}{8}$
shortest line	sixteenths: $\frac{1}{16}$, $\frac{3}{16}$, $\frac{5}{16}$, ..., $\frac{15}{16}$

To name any line, **count the small spaces** from the last whole inch. Each space is $\frac{1}{16}$ in. Then write the fraction in **simplest form**:
$\frac{8}{16} = \frac{1}{2}$ $\frac{12}{16} = \frac{3}{4}$ $\frac{6}{16} = \frac{3}{8}$

Calculator. The TI-30XIIS simplifies for you: 6 [A b/c] 16 [ENTER] shows 3⌟8, which means $\frac{3}{8}$.

Example 2

(a) This is **one inch**, made bigger. Write the fraction for each line, in simplest form, in the boxes. The half is done for you.



(b) Write in simplest form. $\frac{2}{16} =$ _____ $\frac{4}{16} =$ _____ $\frac{10}{16} =$ _____ $\frac{14}{16} =$ _____

(c) Which lines can **not** be simplified? How can you tell them on a ruler?

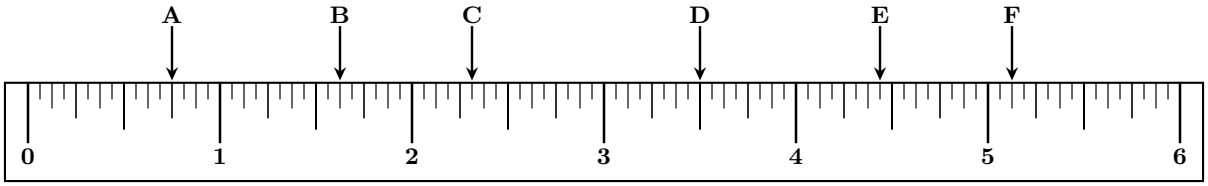
3. Reading a Ruler

Reading a ruler or tape measure
Step 1. The **whole inches** are the last number you passed.
Step 2. The **fraction** is how far past that number: use the length of the line, or count the $\frac{1}{16}$ spaces.
Step 3. Write a mixed number in simplest form, with the unit: $4\frac{7}{16}$ in (say "four and seven-sixteenths inches").

Watch out. Line up the end of the object with the **0 line**, not with the end of the ruler. Many rulers have a little blank space before 0. If the object does not start at 0, you must take away the starting mark.

Example 3

Read the length at each arrow. This ruler is real size.

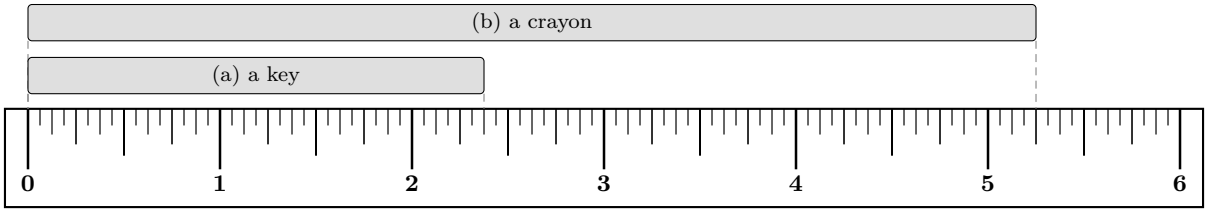


A _____ B _____ C _____

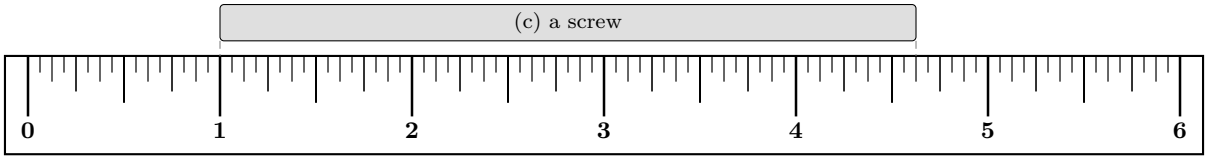
D _____ E _____ F _____

Example 4

How long is each object? The rulers are real size.



(a) the key _____ (b) the crayon _____



(c) The screw starts at the 1 in line. How long is it?

4. Measuring to the Nearest $\frac{1}{8}$ or $\frac{1}{16}$ Inch

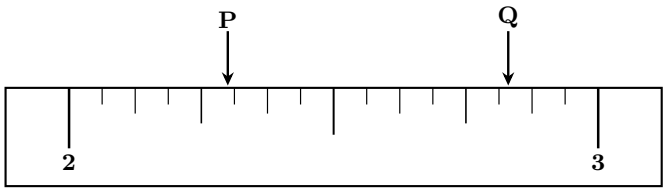
An edge often falls **between** two lines. Then you use the **closest** line.

- **To the nearest $\frac{1}{16}$ in:** use all the lines, and pick the closest one.
- **To the nearest $\frac{1}{8}$ in:** skip the shortest ($\frac{1}{16}$) lines. Pick the closest of the longer lines.

A rough job (framing a wall) is often fine to the nearest $\frac{1}{8}$ in. Finish work (trim, cabinets) needs the nearest $\frac{1}{16}$ in. Using smaller marks gives a more **precise** measurement.

Example 5

This is the part of a ruler from 2 in to 3 in, made bigger. Measure each arrow.



	nearest $\frac{1}{16}$ in	**nearest $\frac{1}{8}$ in**
P		
Q		

5. Tape Measures and Feet and Inches

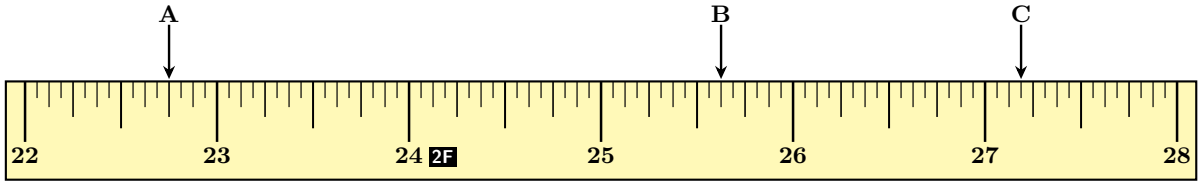
A **tape measure** is a long ruler that rolls up. Its **hook** at the end is a little loose on purpose. It slides by its own thickness, so the reading is right when you hook it over an edge and when you push it against a wall.

The numbers on a tape keep counting inches past 12: 13, 14, 15, … At each foot many tapes add a small tag: **1F** at 12 in, **2F** at 24 in, **3F** at 36 in. Most tapes also show every 16 in in **red** (16, 32, 48, …) because wall studs are built 16 in apart.

Feet and inches: 2 feet $1\frac{5}{8}$ inches $= 2'\,1\frac{5}{8}'' = 2$ ft $1\frac{5}{8}$ in. The inches part is always **less than 12**.
On a tape: the **feet** are the last foot tag you passed (1F, 2F, …). The **inches** are how far past that tag.
e.g. $25\frac{5}{8}$ in is $1\frac{5}{8}$ in past 2F (24 in), so it is $2'\,1\frac{5}{8}''$. (Lesson 4 shows how to change any number of inches.)

Example 6

Read each arrow on this part of a tape measure (real size). Write it in inches, then in feet and inches.



	inches	**feet and inches**	**how**
A			
B			
C			

Good to know. A "2 by 4" is not 2 in by 4 in. The board shrinks as it dries and is planed smooth, so it is really $1\frac{1}{2}''$ by $3\frac{1}{2}''$.

Example 7

(a) Write in symbols: 6 feet 1 inch and 5 feet $10\frac{1}{2}$ inches. Write in words: $8'$

(b) Order these volleyball players from tallest to shortest.
Kenji $5'\,11''$ Ruth $6'\,2\frac{1}{2}''$ Marc $6'\,0''$ Ines $5'\,9\frac{3}{4}''$

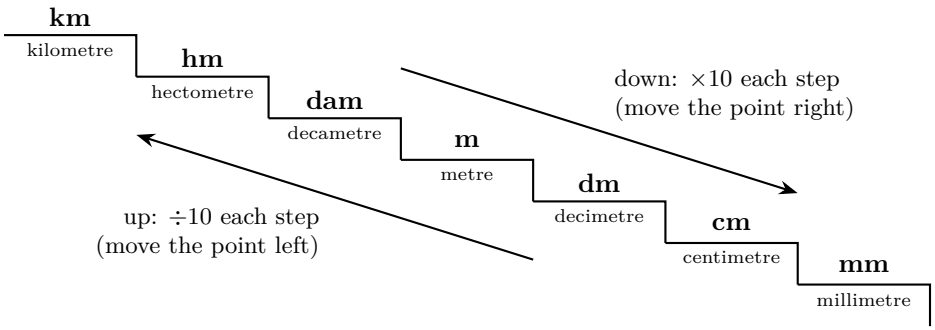
(c) A mechanic's wrenches are marked $\frac{1}{2}''$, $\frac{7}{16}''$, $\frac{5}{8}''$, $\frac{9}{16}''$ and $\frac{3}{4}''$. Put them in order from smallest to largest. The $\frac{9}{16}''$ wrench is too small for a bolt. Which wrench should she try next?

Lesson 4: Converting Length Within SI and Within Imperial

Moving along the SI staircase, using 12, 3 and 5280 for imperial units, and adding and subtracting feet and inches.

1. Converting Within SI: The Staircase

Every SI unit of length is the **metre** with a **prefix** in front. On the staircase, each step is **10 times** as big as the step below it. To **convert** (change) a length to another unit, count the steps.



Going down to a smaller unit: **multiply** by 10 for each step. **Going up** to a bigger unit: **divide** by 10 for each step.

1 km = 1000 m 1 m = 100 cm 1 m = 1000 mm 1 cm = 10 mm

Watch out. A **smaller** unit means a **bigger** number: 4.2 m is 420 cm, not 0.042 cm. Before you finish, ask: "Should my answer be bigger or smaller than what I started with?"

Good to know. People mostly use km, m, cm and mm. The hm, dam and dm steps are rarely used, but they still count when you count steps (m to mm is 3 steps).

Example 1

Convert each length. Fill in the table.

Convert	**Up or down? Steps?**	**× or ÷ by**	**Answer**
(a) 4.2 m to cm			
(b) 85 mm to cm			
(c) 2.75 km to m			
(d) 630 cm to m			
(e) 0.6 m to mm			
(f) 12 500 m to km			
(g) 45 mm to m			

Mixed units. A length like 3 m 45 cm uses two units. You can only add, subtract or compare lengths when they are in the **same unit**, so convert first.

Example 2

(a) Write 3 m 45 cm in centimetres, and in metres.

(b) A plumber cuts 65 cm off a 2.4 m length of pipe. How long is the piece that is left, in cm and in m?

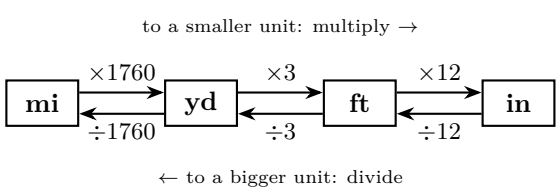
(c) Put in order from shortest to longest:
1.2 km, 950 m, 125 000 cm, 1 300 000 mm

(d) Maya runs 4 laps of a 400 m track, then 1.8 km home on the road. How far does she run in all, in km?

2. Converting Within Imperial: Multiply or Divide?

The imperial units of length are the **inch** (in or ″), the **foot** (ft or ′), the **yard** (yd) and the **mile** (mi). They do **not** go up by tens, so you need to know these numbers.

12 in = 1 ft 3 ft = 1 yd 36 in = 1 yd
5280 ft = 1 mi 1760 yd = 1 mi
Bigger unit to smaller unit: multiply.
Smaller unit to bigger unit: divide.



Unit factors. A **unit factor** is a fraction that equals 1, such as $\frac{12 \text{ in}}{1 \text{ ft}}$. Multiplying by it does not change the length,

only the unit. Put the unit you want on **top** and the unit you have on the **bottom**, so the old unit cancels:

$$7\cancel{\text{ft}} \times \frac{12 \text{ in}}{1\cancel{\text{ft}}} = 7 \times 12 = 84 \text{ in} \qquad 54\cancel{\text{in}} \times \frac{1 \text{ ft}}{12\cancel{\text{in}}} = 54 \div 12 = 4.5 \text{ ft}$$

Unit factors are handy when you need two steps (miles to feet to inches): just multiply by one factor after another.

Example 3

Convert each length. Fill in the table.

Convert	**To a bigger or smaller unit?**	**Work**	**Answer**
(a) 9 ft to in			
(b) 66 in to ft			
(c) 15 ft to yd			
(d) 4.5 yd to ft			
(e) 2 mi to ft			
(f) 7920 ft to mi			

Example 4

(a) A football field is 100 yd long between the goal lines. How long is that in inches? Use unit factors.

(b) A spool holds 500 ft of electrical wire. An electrician uses 84 in of wire for each light fixture. How many fixtures can she wire from one spool?

3. Feet and Inches

Heights and building measurements are often written in **feet and inches**, like $5'\,7''$ (say "5 feet 7 inches"). The same length can be written three ways:

feet and inches: $5'\,6''$ inches only: 66 in **decimal feet**: 5.5 ft

ft and in → in: feet × 12, then add the inches.	$5'\,7'' = 5 \times 12 + 7 = 67$ in
in → ft and in: ÷ 12; the whole number is the feet;	100 in: $100 \div 12 = 8.33\ldots$, so 8 ft;
inches left = inches − feet × 12.	$100 - 8 \times 12 = 4$, so $8'\,4''$
ft and in → decimal ft: inches ÷ 12, add to the feet.	$6'\,9'' = 6 + 9 \div 12 = 6.75$ ft
decimal ft → ft and in: decimal part × 12.	4.25 ft: $0.25 \times 12 = 3$, so $4'\,3''$

Watch out. 5.6 ft is **not** $5'\,6''$. The 0.6 is a part of a *foot*, not inches: $0.6 \times 12 = 7.2$, so 5.6 ft $\approx 5'\,7''$. (And $5'\,6'' = 5.5$ ft.)

Handy facts. 3 in = 0.25 ft 6 in = 0.5 ft 9 in = 0.75 ft 4 in ≈ 0.33 ft 8 in ≈ 0.67 ft

Calculator. For 100 in: 100 [÷] 12 [ENTER] shows 8.333333, so 8 ft. Then 100 [−] 8 [×] 12 [ENTER] shows 4, so 4 in left.

Example 5

Convert each length. Round decimal feet to the nearest hundredth and inches to the nearest inch.

Length	Change to	Work	Answer
(a) $6'\,2''$	inches		
(b) $5'\,10\frac{1}{2}''$	inches		
(c) 45 in	ft and in		
(d) 130 in	ft and in		
(e) $8'\,6''$	decimal ft		
(f) $3'\,4''$	decimal ft		
(g) 7.25 ft	ft and in		
(h) 5.6 ft	ft and in		

Example 6

(a) Last year Jayden was $4'\,11''$ tall. Now he is 62 in tall. How much did he grow?

(b) A laser measure shows a wall is 9.75 ft long. Write this in feet and inches.

(c) Which is longer, 7.3 ft or $7'\,3''$? By how much?

4. Adding and Subtracting Feet and Inches

Add **feet to feet** and **inches to inches**. If you get 12 or more inches, trade 12 in for 1 ft (**regroup**), like trading 10 ones for 1 ten. When you subtract and the top has too few inches, **borrow** 1 ft and turn it into 12 in. To multiply, multiply the feet and the inches, then regroup.

Add: $4'\,7'' + 3'\,10'' = 7'\,17''$

$17'' = 12'' + 5'' = 1'\,5''$, so $7'\,17'' = 8'\,5''$

Subtract: $9'\,3'' - 5'\,8''$

borrow 1 ft: $9'\,3'' = 8'\,15''$; $8'\,15'' - 5'\,8'' = 3'\,7''$

Check. Change everything to inches. For the addition: $55 + 46 = 101$ in, and $101 - 96 = 5$, so $8'\,5''$ ✓

Example 7

Find each answer. Write it in feet and inches, with fewer than 12 inches.

(a) $5'\,9'' + 2'\,8''$

(b) $11'\,2'' - 6'\,7''$

(c) $3 \times 2'\,9''$

(d) $6'\,4\frac{1}{2}'' + 2'\,9\frac{3}{4}''$

(e) $8' - 3'\,5\frac{1}{2}''$

Example 8

Owen's garage door opening is $7'\,0''$ high. His SUV is $5'\,11''$ tall.

(a) He puts a cargo box on the roof that adds $1'\,2''$. Will the SUV fit through the door?

(b) A slimmer cargo box adds only $11\frac{1}{2}''$. How much room is left above it?

5. Cutting and Fitting

Lumber is sold in lengths such as 8 ft, 10 ft and 12 ft. Trim and pipe are often sold in 8 ft or 2.4 m pieces. To plan the cuts:

- **Step 1.** Change everything to the same, smaller unit (inches, or cm).
- **Step 2.** Divide the board length by the piece length.
- **Step 3.** Round in the way that makes sense.

Pieces you can **cut** from one board: **round down** (a short leftover is not a piece). Boards, rolls or packs you must **buy**: **round up** (you cannot buy part of a board).

On the job. Each saw cut turns about $\frac{1}{8}''$ of wood into sawdust (the **kerf**). For tight jobs, leave room for it.

Example 9

Noah is building garden planters. He needs pieces of board 30 in long. The boards are 8 ft long and cost $5.48 each.

(a) How many 30 in pieces can he cut from one board? How much is left over?

(b) He needs 14 pieces. How many boards must he buy, and what do they cost?

(c) For the planter trim, he buys trim in 2.4 m lengths and cuts it into 55 cm pieces. How many pieces can he cut from one length, and how much is left over?

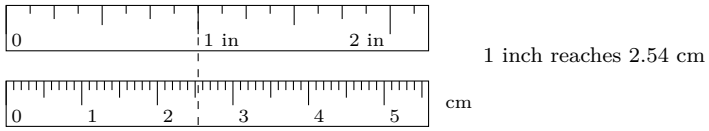
Lesson 5: Converting Length Between SI and Imperial

Changing inches, feet, yards and miles to centimetres, metres and kilometres (and back), estimating in your head, and checking by converting back.

1. The Conversion Facts and Quick Estimates

Canada uses the SI (metric) system, but imperial units are still all around us: lumber and plywood, people's heights, TV and tire sizes, and every road sign in the United States. To change a length from one system to the other, you use a **conversion fact**. The inch fact is **exact**, and so are the foot and yard facts, which are built from it. The others are rounded, so we write ≈ ("is about").

Conversion fact	**Quick estimate**	**Referent**
1 in = 2.54 cm = 25.4 mm (exact)	about 2.5 cm	the width of your thumb
1 ft = 30.48 cm	about 30 cm	a 30 cm ruler is just under 1 ft
1 yd = 0.9144 m	a bit less than 1 m	one long step
1 m ≈ 3.28 ft ≈ 39.37 in	a bit more than 3 ft	a metre stick is about 3 in longer than a yardstick
1 mi ≈ 1.609 km	about 1.6 km	about 20 minutes of walking
1 km ≈ 0.621 mi	about 0.6 mi	a bit more than half a mile



Quick estimates in your head. Use the rounded facts.

in → cm: $\times 2.5$ (double it, then add half of the number)	e.g. 12 in: $24 + 6 = 30$ cm
cm → in: $\div 2.5$ (or $\times 4$, then $\div 10$)	e.g. 50 cm: $50 \times 4 = 200$, $200 \div 10 = 20$ in
mi → km: $\times 1.6$ (add six tenths of the number)	e.g. 50 mi: $50 + 30 = 80$ km
km → mi: $\times 0.6$ (six tenths of the number)	e.g. 100 km: about 60 mi

Example 1

(a) Which length is longer? Use the conversion facts.

Pair	Longer	How I know
(i) 1 in or 1 cm		
(ii) 1 yd or 1 m		
(iii) 1 mi or 1 km		
(iv) 1 ft or 25 cm		
(v) 10 in or 25 cm		

(b) Estimate in your head. Show your thinking.

(i) A TV screen measures 55 in across. About how many centimetres is that?

(ii) A highway sign says "Edmonton 300 km". About how many miles is that?

(iii) A hiking trail in Montana is 5 mi long. About how many kilometres is that?

2. Converting with a Unit Factor or a Proportion

For an exact answer, multiply by a **unit factor**. A unit factor is a fraction made from a conversion fact, like $\frac{2.54 \text{ cm}}{1 \text{ in}}$. Its top and bottom are the same length, so it is equal to 1. Multiplying by it changes the unit but not the length.

Step 1. Write the length you have, with its unit.
Step 2. Multiply by a unit factor. Put the unit you **have** on the **bottom** and the unit you **want** on the **top**.
Step 3. Cross out the unit that is on the top and the bottom. Calculate, then round.
Step 4. Check with a quick estimate, or convert your answer back.

For example, 9 in to centimetres: $9 \cancel{\text{in}} \times \frac{2.54 \text{ cm}}{1 \cancel{\text{in}}} = 9 \times 2.54 = 22.86$ cm. Check: $9 \times 2.5 = 22.5$, close.

And 60 cm to inches: $60 \cancel{\text{cm}} \times \frac{1 \text{ in}}{2.54 \cancel{\text{cm}}} = 60 \div 2.54 \approx 23.6$ in. (The 2.54 is on the bottom, so you **divide**.)

You can also use a **proportion**, as in Unit 1. Keep the same units in the same places: $\frac{1 \text{ in}}{2.54 \text{ cm}} = \frac{9 \text{ in}}{x}$ gives $x = 9 \times 2.54 = 22.86$ cm.

Watch out. Ask: should the number get bigger or smaller? A centimetre is **shorter** than an inch, so it takes **more** centimetres: 9 in is about 23 cm, not about 3.5 cm. In the same way, a kilometre is shorter than a mile, so a distance in km is a bigger number than the same distance in miles.

Calculator. 9 [×] 2.54 [ENTER] shows 22.86. 60 [÷] 2.54 [ENTER] shows 23.62204..., about 23.6.

If a fact gives centimetres but you want metres, convert inside SI at the end: 182.88 cm = $182.88 \div 100 = 1.8288$ m.

Example 2

Convert. Round to the nearest tenth (to the nearest hundredth in (d)).

Convert	Work (unit factor)	Answer
(a) 9 in to cm		
(b) 60 cm to in		
(c) 6 ft to cm		
(d) 6 ft to m		
(e) 50 m to yd		
(f) 13.1 mi to km		
(g) 5 km to mi		

Fractions of an inch. Change the fraction to a decimal first: $3\frac{1}{2}$ in $= 3.5$ in, $\frac{5}{16}$ in $= 5 \div 16 = 0.3125$ in. Then convert.

Example 3

A "2 by 4" from the lumber store is not really 2 in by 4 in. After it is dried and planed smooth, it is $1\frac{1}{2}$ in by $3\frac{1}{2}$ in.

(a) Use a proportion to change $3\frac{1}{2}$ in to centimetres.

(b) Change $1\frac{1}{2}$ in to millimetres.

(c) Check your answer to (a) by converting it back to inches.

(d) Ravi's metric plan needs a board at least 9 cm wide. Is the wide side of a 2 by 4 wide enough? Explain.

3. Heights and Lengths in Feet and Inches

A height like $5'\,10''$ (5 ft 10 in) uses two units. Change it to **one** unit (inches) before you convert.

Feet and inches → cm: Step 1: inches = feet × 12 + inches. Step 2: × 2.54.

cm → feet and inches: Step 1: ÷ 2.54 to get inches (round). Step 2: take out as many 12s as you can. They are the feet. What is left over is the inches.

Watch out. $5'\,10''$ is **not** 5.10 ft. There are 12 inches in a foot, not 10. On a calculator, $70 \div 12 = 5.833\ldots$, and the 0.833 is part of a foot, not 83 inches. Instead: $5 \times 12 = 60$, $70 - 60 = 10$, so 70 in $= 5'\,10''$.

Example 4

(a) Change each height to centimetres. Round to the nearest centimetre.

Person	Height	Work	Height in cm
Marcus	$5' 10''$		
Priya	$5' 2''$		
Owen (age 5)	$3' 6''$		

(b) Hana's driver's licence says her height is 168 cm. Find her height in feet and inches, to the nearest inch.

Example 5

The entrance to a parkade has a sign: "Clearance $6' 8''$". Devon's SUV with a cargo box on the roof is 2.05 m tall.

(a) Change $6' 8''$ to metres.

(b) Will Devon's SUV fit? By how much?

(c) Without the cargo box, the SUV is 1.78 m tall. How much space is there above it, in centimetres?

Metric lengths on an inch tape. An inch tape measure shows sixteenths of an inch. To find a metric length on it, change to inches, then change the decimal part to **sixteenths**.

Decimal inches → nearest $\frac{1}{16}$ in: multiply the decimal part by 16, round to a whole number, write it over 16 and simplify.
e.g. 2.3 in: $0.3 \times 16 = 4.8 \approx 5$, so 2.3 in $\approx 2\frac{5}{16}$ in.

Example 6

(a) A drill bit is $\frac{5}{16}$ in wide. What is that in millimetres, to the nearest tenth?

(b) A metric plan says a shelf is 45 cm deep. Kim has an inch tape. Find 45 cm to the nearest $\frac{1}{16}$ in.

(c) Check (b) by converting $17\frac{11}{16}$ in back to centimetres.

4. Distance, Speed and Price per Length

Road signs in the United States use **miles** and speeds in **miles per hour** (mph). Canadian signs use km and km/h. To convert a speed, use the same fact as for distance; the "per hour" does not change: 60 mph $= 60 \times 1.609 \approx 96.5$ km/h.

Example 7

The Tran family drives from Calgary to Montana. Their car's speedometer and odometer show only km/h and km. Round to the nearest whole number.

(a) The speed limit on the interstate highway is 80 mph. What is that in km/h?

(b) A sign says "Great Falls 87 mi". How far is that in kilometres?

(c) Near a town, the limit drops to 55 mph. Mr. Tran is driving at 95 km/h. Is he speeding? By how much?

(d) At the sign in (b), the odometer read 48 312 km. What should it read at Great Falls?

Price per length. Fabric, rope, trim and chain are often sold by the foot or by the yard. To compare with a price per metre, find the price of **one metre** (a unit price, as in Unit 1).

Price per metre = price ÷ length in metres
e.g. \$12.49 per yd: 1 yd = 0.9144 m, so the price per metre is \$12.49 ÷ 0.9144 ≈ \$13.66
(For a price per foot, divide by 0.3048, because 1 ft = 30.48 cm = 0.3048 m.)

Round up when you need enough. If you need at least a certain length, and the store sells it in steps (whole yards, quarter yards), **round up** to the next step. 2.73 yd means you must buy $2\frac{3}{4}$ yd: $2\frac{1}{2}$ yd would be too short.

Example 8

Mei is making curtains. Fabric Barn sells her fabric for \$12.49 per yard, cut in steps of $\frac{1}{4}$ yd. Sew Central sells the same fabric for \$13.29 per metre, cut to any length.

(a) Find Fabric Barn's price per metre. Which store has the lower price per metre?

(b) Mei needs 2.5 m. How many yards must she buy at Fabric Barn, and what does it cost?

(c) What does 2.5 m cost at Sew Central? How much does Mei save there?

Lesson 6: Perimeter and Circumference

Adding the sides of a shape, finding missing sides, finding the distance around a circle, and pricing fencing, trim and edging.

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1. Perimeter of a Polygon

A **polygon** is a flat shape with straight sides, like a triangle, a rectangle or a hexagon. The **perimeter** is the total distance **around the outside** of a shape. To find it, add the lengths of all the sides. Perimeter is a **length**, so its unit is cm, m, in or ft (no small 2).

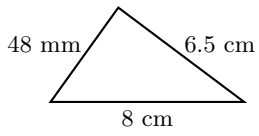
Perimeter P = the sum of all the side lengths	
Square (side s): $P = 4s$	**Rectangle** (length l, width w): $P = 2l + 2w$ or $P = 2(l + w)$

Watch out. Start at one corner and go around the shape. Tick each side as you add it, so you do not miss one or count one twice. All the sides must be in the **same unit** before you add.

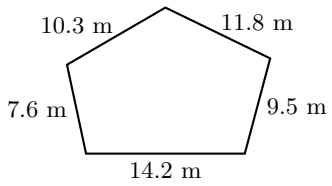
Reading diagrams. A small square in a corner means a **right angle** (a square corner, 90°). Small tick marks on sides mean those sides are **equal**. Diagrams are often not drawn to scale: use the numbers, not a ruler.

Example 1

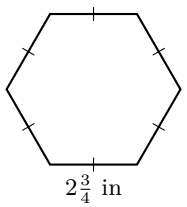
Find the perimeter of each shape.



(a) a triangle



(b) a building lot (not to scale)



(c) a hexagon tile (all sides equal)

Example 2

Find the perimeter of each rectangle or square. Estimate first to check your answer.

Shape	Size	Work	Perimeter
(a) Volleyball court	18 m by 9 m		
(b) Phone screen	15.2 cm by 7.1 cm		
(c) Square floor tile	side $4\frac{3}{4}$ in		
(d) Area rug	5 ft 6 in by 7 ft 9 in		

2. Finding Missing Side Lengths

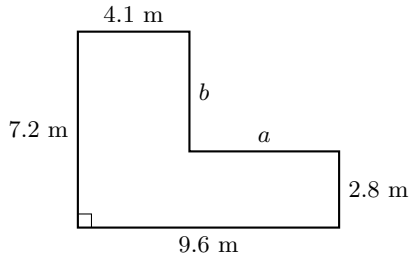
Sometimes a diagram does not show every side. When all the corners are right angles (like an L shape), use the sides you know:

- The sides that go **across** at the top add up to the side across the bottom.
- The sides that go **up and down** on one side add up to the side on the other.

Missing side = the long side − the part you know	Then add **all** the sides, including the ones you found.

Check. When a shape is a rectangle with **one corner cut out** (an L shape), its perimeter is the same as the perimeter of the whole rectangle around it. The cut just moves two sides inward. (This shortcut does **not** work when a notch is cut into the middle of a side.)

Example 3



All corners are right angles.

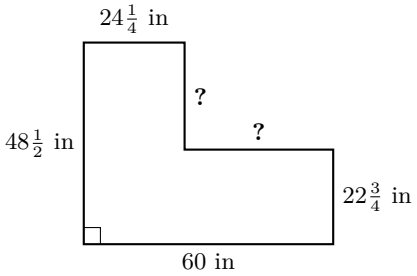
Rosa is planting a garden shaped like an L. She wants to put a short border fence around it.

(a) Find the missing side lengths a and b.

(b) Find the perimeter of the garden.

(c) Check with the shortcut for an L shape.

Example 4



All corners are right angles.

Tyler builds an L-shaped desk top. He will glue a thin strip of edge banding all the way around it.

(a) Find the two missing side lengths.

(b) How many inches of edge banding does Tyler need?

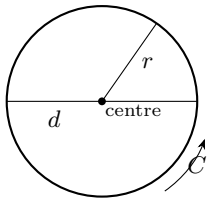
(c) Write your answer in feet and inches.

3. Circumference of a Circle

A circle has no straight sides, so we need new words.

- The **centre** is the point in the middle.
- The **radius** (r) is the distance from the centre to the edge.
- The **diameter** (d) is the distance all the way across, through the centre. It is two radii: $d = 2r$.
- The **circumference** (C) is the distance around the circle. It is the perimeter of a circle.

For **every** circle, the circumference is a little more than 3 times the diameter. The exact number is called **pi** (say "pie"), written π. It is about 3.14, and its decimals go on forever.



Circumference: $C = \pi d$ or $C = 2\pi r$ ($\pi \approx 3.14$)	
Estimate: $C \approx 3 \times d$ e.g. a circle 30 cm across goes around a bit more than $3 \times 30 = 90$ cm.	

Calculator. Use the [π] key: it holds many more decimals than 3.14. [π] [×] 30 [ENTER] shows 94.24777961. With a radius of 7: 2 [×] [π] [×] 7 [ENTER] shows 43.98229715. (Using 3.14 gives an answer that is a tiny bit different.)

Watch out. If you are given the **radius**, double it first (or use $C = 2\pi r$). The most common mistake is $\pi \times r$, which gives only half the circumference.

Example 5

Find the circumference of each object. Estimate first. Round to the nearest tenth.

Object	**Given**	**Diameter**	**Estimate** ($3 \times d$)	**Work** ($C = \pi d$)	**Circumference**
(a) Wall clock	$d = 30$ cm				
(b) Pizza	$r = 7$ in				
(c) Round hay bale	$r = 0.9$ m				
(d) Jar lid	$d = 3\frac{1}{4}$ in				

Example 6

Each time a bike wheel turns once, the bike rolls forward a distance equal to the wheel's circumference. Nadia's bike wheel has a diameter of 66 cm.

(a) How far does the bike roll when the wheel turns once? Round to the nearest tenth of a centimetre.

(b) How far does the bike roll in 500 turns? Give your answer in metres, to the nearest tenth.

(c) About how many turns does the wheel make in 1 km?

4. Shapes With Half Circles

A **semicircle** is half of a circle. Its curved edge is half of the circumference. Two semicircles with the same diameter make one whole circle.

Many shapes are made of a rectangle and one or two semicircles (an arched window, a running track). To find the perimeter, add only the edges on the **outside**. Do not add a line where two pieces join.

Curved edge of a semicircle $= \pi d \div 2$	**Two equal semicircles** $= \pi d$ (one whole circle)
Perimeter = the straight edges on the outside + the curved edges	

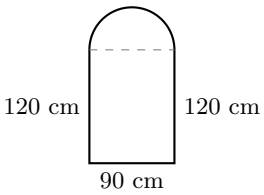
Watch out. A half circle on its own has a straight edge too (its diameter). Remember to add it. But when the half circle sits on a rectangle, that line is **inside** the shape, so it is not part of the perimeter.

Example 7

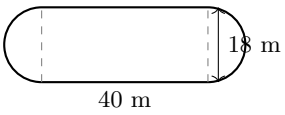
Find the perimeter of each shape. Round to the nearest tenth.



(a) a half-circle rug



(b) an arched window frame



(c) an outdoor skating path: two straight sides 40 m long, and a semicircle at each end

5. Fencing, Trim and Edging: Finding the Cost

Fencing a yard, putting baseboard in a room and edging a garden are all perimeter jobs. Follow these steps.

Step 1. Find the length needed. Use only the sides that need it, and subtract any gaps (doors, gates).

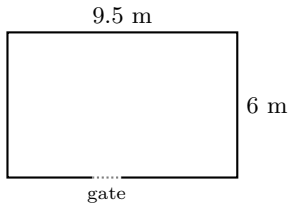
Step 2. If it is sold **by the metre or foot**: cost = length × price per metre (or per foot).

Step 3. If it is sold in **pieces or rolls of a fixed length**: divide to find how many, then **round up**. You cannot buy part of a piece. Cost = number of pieces × price per piece.

Cost = length × price per unit length **Pieces needed** = length needed ÷ length of one piece → **always round up**

Watch out. Round pieces **up** even when the decimal is small: 5.1 pieces means you must buy 6. If you buy 5, you will be short.

Example 8



Jamal is fencing a rectangular dog run, 9.5 m by 6 m. He will leave a 1.2 m space for a gate. Fencing costs $18.75 per metre, and the gate costs $129.00.

(a) Find the perimeter of the dog run.

(b) How many metres of fencing does Jamal need?

(c) Find the cost of the fencing.

(d) Find the total cost of the fencing and the gate, with 5% GST.

Example 9

Shawna is putting new baseboard around her bedroom. The room is a rectangle 11 ft 6 in by 13 ft 3 in. There is no baseboard across the 2 ft 8 in doorway. Baseboard is sold in 8 ft pieces for $13.98 each. (Pieces can be cut and joined anywhere.)

(a) Find the perimeter of the room in feet and inches.

(b) How much baseboard does Shawna need?

(c) How many 8 ft pieces must she buy, and what do they cost?

(d) How much baseboard will be left over?

Lesson 7: Rearranging Formulas: Finding a Missing Side

Use the perimeter, circumference and area formulas backwards to find a missing side, diameter or radius, then check by putting your answer back in.

1. Formulas, Variables and Undoing

A **formula** is a rule written with letters. Each letter is a **variable**: it stands for a number. To use a formula, you **substitute**: you replace each letter you know with its number. Usually you know every value but one, and you want to find the missing one.

Formulas in this lesson

Rectangle: $P = 2l + 2w$ (l = length, w = width) Square: $P = 4s$ Triangle: $P = a + b + c$

Circle: $C = \pi d$ or $C = 2\pi r$ (d = diameter, r = radius) Area of a rectangle: $A = lw$

When the missing value is not by itself, you **undo** what the formula did to it. Each operation has an **opposite** that undoes it:

Opposites: add ↔ subtract multiply ↔ divide

Balance rule: an equation is like a balance. Whatever you do to one side, do the same to the other side.

Example 1

Substitute the numbers you know. Then undo to find the missing value.

(a) A square patio has a perimeter of 46 ft. How long is each side? Use $P = 4s$.

(b) A rectangular rug has an area of 7.5 m^2. It is 3 m long. How wide is it? Use $A = lw$.

(c) Wood edging for a triangle-shaped flower bed is 10 m long. Two sides are 2.8 m and 3.5 m. How long is the third side?

2. One Step: Finding a Diameter, Radius or Width

Sometimes you will use the same formula many times to find the same thing. Then it is faster to **rearrange** the formula first. That means you write it a new way, so the thing you want to find is by itself on one side of the equal sign. When the formula only multiplies, rearranging takes one step: divide.

One-step rearranging

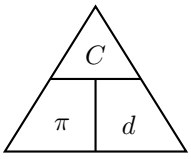
$C = \pi d$ divide both sides by π: $d = C \div \pi$

$C = 2\pi r$ divide both sides by 2π: $r = C \div (2\pi)$ (or $r = d \div 2$)

$A = lw$ divide both sides by l: $w = A \div l$ (and $l = A \div w$)

$P = 4s$ divide both sides by 4: $s = P \div 4$

Check: put your answer back into the *first* formula. You should get the number you started with.



Formula triangle (like pay, rate and hours in Unit 2). Cover d: you see C over π, so $d = C \div \pi$.

Estimate. π is a little more than 3, so a diameter is a little *less* than one third of the circumference. A circumference of 90 cm means a diameter of a bit under 30 cm.

Calculator. For $r = C \div (2\pi)$, use brackets: 113 [÷] [(] 2 [×] [π] [)] [ENTER]. Or divide twice: 113 [÷] 2 [÷] [π] [ENTER]. Typing 113 [÷] 2 [×] [π] is wrong: it multiplies by π.

Example 2

(a) An arborist (a tree expert) wraps a tape measure around a spruce trunk. The circumference is 188 cm. Find the diameter of the trunk, to the nearest tenth of a centimetre.

(b) A round table has 113 in of metal trim around its edge. Find the radius of the table, to the nearest tenth of an inch.

Example 3

(a) Kai's rectangular bedroom floor has an area of 132 ft^2. The room is 12 ft long. How wide is it?

(b) Rearrange $A = lw$ to find the length. Then use it: a rectangular patio uses 22.5 m^2 of paving stones and is 3.6 m wide. How long is it?

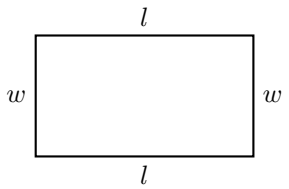
3. Two Steps: The Perimeter of a Rectangle

In $P = 2l + 2w$, the formula does two things to w: it **multiplies** by 2, then it **adds** $2l$. To undo, go in reverse order, like taking off your shoes before your socks.

Rearranging in two steps

Step 1: Undo the + or − first. **Step 2:** Then undo the × or ÷.

$P = 2l + 2w$	
$P - 2l = 2w$	Step 1: subtract $2l$ from both sides.
$(P - 2l) \div 2 = w$	Step 2: divide both sides by 2.
So $w = (P - 2l) \div 2$	and in the same way $l = (P - 2w) \div 2$



Two lengths and two widths go all the way around.

Another way. Half of the perimeter is one length plus one width. So you can also find $w = P \div 2 - l$. For $P = 36$ and $l = 11$: $36 \div 2 - 11 = 18 - 11 = 7$. Same answer.

Watch out. Subtract **two** lengths ($2l$), not one. And don't forget Step 2: $P - 2l$ is the two widths together, not one width.

Example 4

Ava has 36 m of fencing for a rectangular dog run. She wants it to be 11 m long. How wide will it be?

(a) Solve by substituting first.

(b) Solve by rearranging first. Check your answer.

Substitute first or rearrange first. Both ways give the same answer. Substituting first is often easier for one question. Rearranging first is faster when you must do the same thing many times, like filling in a table.

Example 5

Students in the woodshop are building rectangular planter boxes. Each box gets a strip of trim all the way around its top edge.

(a) Rearrange $P = 2l + 2w$ to find the length l. Show both steps.

(b) Use your formula to complete the table.

	Trim (P)	**Width (w)**	**Work:** $(P - 2w) \div 2$	**Length (l)**
Box 1	20 ft	4 ft		
Box 2	11 m	1.5 m		
Box 3	15 ft	$2\frac{3}{4}$ ft		

4. Check Your Answer

Always check a missing side in two ways:

1. Does it make sense? A width is less than half the perimeter. A diameter is about a third of the circumference. A length can't be negative.

2. Substitute back. Put your answer into the *original* formula. You should get the number you started with.

Example 6

Each student made a mistake. Check their answer, explain the mistake, and find the correct answer.

(a) A rectangle has a perimeter of 50 cm and a length of 15 cm. Noah wrote: $w = 50 - 2(15) = 20$ cm.

(b) A round wading pool has a circumference of 94.2 in. Mei wrote: $d = 94.2 \times \pi = 295.9$ in.

Example 7

Maya has \$78.00 to spend on garden edging. The edging costs \$3.25 per foot. She wants to put it all the way around a rectangular garden bed that is 7 ft long. How wide can the bed be?

Lesson 8: Area of 2-D Shapes

Measuring the surface inside a flat shape in square units, using the formulas for rectangles, parallelograms, triangles, trapezoids and circles, and changing between area units to find the cost of a job.

1. What Is Area?

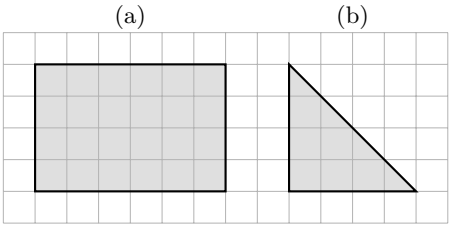
The **area** of a flat shape is the amount of surface inside it. We measure area by counting **square units**. A square that is 1 cm long and 1 cm wide has an area of **1 square centimetre**, written **1 cm^2**.

Other square units: square metres (m^2), square inches (in^2), square feet (ft^2) and square yards (yd^2). Some **referents**: your fingernail is about 1 cm^2; a 1 ft by 1 ft floor tile is 1 ft^2; a big square of cardboard 1 m by 1 m is 1 m^2.

Area vs perimeter. Perimeter is the distance *around* a shape (fencing, trim). It is a length, in cm, m, in or ft. **Area** is the surface *inside* (carpet, paint, sod). It is in *square* units.

Example 1

Each small square stands for 1 cm^2. Find the area of each shape by counting squares.



(a) the rectangle _____

(b) the triangle _____

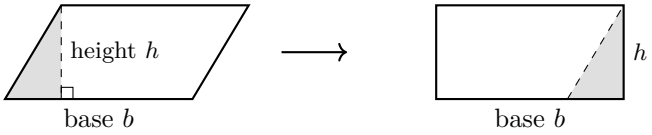
(c) How could you find the area of (a) without counting every square?

_____

Counting works for small shapes on a grid. For real rooms, yards and pieces of fabric, we use a **formula** instead.

2. Rectangles and Parallelograms

A rectangle is rows of squares, so its area is **length times width**. A **parallelogram** is a "pushed over" rectangle with two pairs of parallel sides. Cut a triangle off one end and slide it to the other end: you get a rectangle with the same base and height.



Rectangle: $A = lw$ (length × width) **Square:** $A = s \times s = s^2$

Parallelogram: $A = bh$ (base × height)

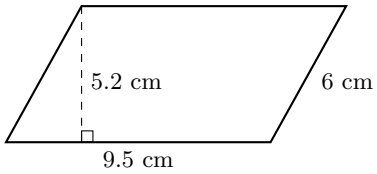
The **height** is measured *straight* across, at a right angle (90°) to the base. It is not the slanted side.

Watch out. Both lengths must be in the **same unit** before you multiply. Metres times metres gives square metres: m × m = m^2.

Example 2

(a) A bedroom floor is a rectangle 4.2 m long and 3.6 m wide. Estimate its area, then find it.

(b) Find the area of the parallelogram. Which length do you *not* use?



Many imperial measurements have fractions. The calculator can work with them for you.

Calculator. To enter $11\frac{1}{4}$, press 11 [A b/c] 1 [A b/c] 4. So $11\frac{1}{4} \times 35\frac{1}{2}$ is 11 [A b/c] 1 [A b/c] 4 [×] 35 [A b/c] 1 [A b/c] 2 [ENTER]. The answer shows as a mixed number, with a `u` between the whole number and the fraction (`2u3_4` means $2\frac{3}{4}$). Press [2nd] [F<>D] to see it as a decimal.

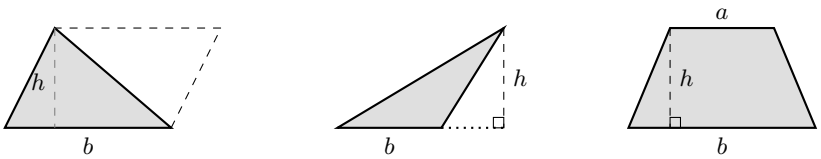
Example 3

A carpenter cuts a shelf from a pine board. The shelf is $35\frac{1}{2}$ in long and $11\frac{1}{4}$ in wide. Find the area of the top of the shelf. Check that your answer is reasonable.

3. Triangles and Trapezoids

Two copies of the same **triangle** fit together to make a parallelogram. So a triangle's area is **half** of base times height. The height goes from the top corner straight down to the base. For some triangles, the height falls *outside* the triangle, so you may need to extend the base with a dotted line.

A **trapezoid** has exactly one pair of parallel sides, a and b. Two copies of a trapezoid also make a parallelogram, with base $a + b$. So its area is half of $(a + b)$ times the height.



triangle = half a parallelogram | height outside the triangle | trapezoid

Triangle: $A = \frac{1}{2}bh$ (or $A = b \times h \div 2$) **Trapezoid:** $A = \frac{1}{2}(a + b)h$ (or $A = (a + b) \times h \div 2$)

a and b are the two **parallel** sides. The height h is the straight distance between them.

Calculator. Use brackets for a trapezoid so the adding is done first: (2.4 + 3.8) × 1.5 ÷ 2 ENTER gives 4.65.

Watch out. Don't forget the $\frac{1}{2}$. Base × height alone gives the area of the whole parallelogram, which is double the triangle.

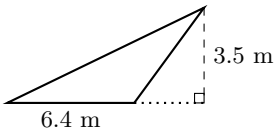
Example 4

Find the area of each triangle.

(a) A triangle with base 8 cm and height 5 cm.

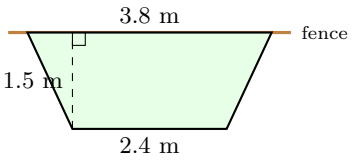
(b) The triangle at the top of a shed wall has base 8 ft and height $3\frac{1}{2}$ ft.

(c)



Example 5

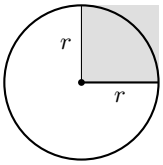
Nadia is building a flower bed along her back fence in Okotoks. It is a trapezoid. The side along the fence is 3.8 m, the front side is 2.4 m, and the two parallel sides are 1.5 m apart.
(a) Find the area of the flower bed.



(b) Garden soil is sold by the bag. One bag covers 0.5 m^2. How many bags should Nadia buy?

4. Circles

The **radius** r of a circle goes from the centre to the edge. The **diameter** d goes all the way across through the centre, so $d = 2r$ and $r = d \div 2$.



The shaded square is r by r, so its area is r^2. The circle's area is a bit more than **3** of these squares. The exact number is π (pi), about 3.14.

Circle: $A = \pi r^2$ (that is, $\pi \times r \times r$)	**Estimate:** $A \approx 3 \times r \times r$

Calculator. Use the [π] key and the [x^2] key. For a radius of 45 cm: [π] [×] 45 [x^2] [ENTER] gives 6361.725...

Watch out. If you are given the **diameter**, halve it first to get the radius. And r^2 means $r \times r$, not $r \times 2$.

Example 6

Round each area to the nearest tenth.

(a) A round table top has a radius of 45 cm. Find its area.

(b) A round rug has a diameter of 8 ft. Estimate its area, then find it.

Example 7

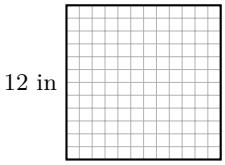
A pizza's size is its diameter. Pizza Palace sells one 16 in pizza for \$21.99, or two 10 in pizzas for \$11.99 each.

(a) Which choice gives more pizza?

(b) Find the price per square inch of each choice, to a tenth of a cent. Which is the better buy?

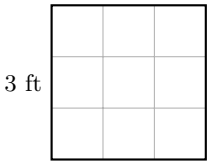
5. Changing Area Units, and the Cost of a Job

A square foot is 12 in by 12 in, so it holds $12 \times 12 = 144$ square inches. A square yard is 3 ft by 3 ft, so it holds $3 \times 3 = 9$ square feet. In the same way, 1 m^2 is 100 cm by 100 cm = 10 000 cm^2.



12 in = 1 ft

1 ft^2 = 144 in^2



3 ft = 1 yd

1 yd^2 = 9 ft^2

Area units 1 ft^2 = 144 in^2 1 yd^2 = 9 ft^2 1 m^2 = 10 000 cm^2 1 $m^2 \approx 10.76$ ft^2
Big unit → small unit: **multiply**. Small unit → big unit: **divide**.
Cost of a job = area × price per square unit

Watch out. 1 ft^2 is *not* 12 in^2. The 12 is used twice (12×12), once for each side. If the lengths are in different units (like ft and in), you can also change the lengths first, then multiply.

Example 8

Complete the table. Round to the nearest tenth if needed.

Change	Work	Answer
(a) 5 ft^2 to in^2		
(b) 2880 in^2 to ft^2		
(c) 27 ft^2 to yd^2		
(d) 3.5 m^2 to cm^2		
(e) 12 m^2 to ft^2		

Materials like flooring, sod and tile are sold in whole **boxes**, **rolls** or **pieces**. Always round the number you buy **up**. Installers also buy about **10% extra** for cutting and mistakes (multiply the area by 1.10).

Example 9

Owen is putting vinyl plank flooring in a bedroom that is 12 ft 6 in long and 10 ft wide. One box covers 23.4 ft^2 and costs \$54.99. He adds 10% for waste.

(a) Find the area of the floor in ft^2.

(b) How much flooring does Owen need with the 10% extra?

(c) How many boxes must he buy?

(d) What is the total cost with 5% GST?

Lesson 9: Composite and Irregular Shapes

Finding the area of shapes made of simple shapes, shapes with a piece cut out, and irregular shapes on a grid, and checking the answer a second way.

1. Composite Shapes: Split and Add

A **composite shape** is a shape made of two or more simple shapes, such as rectangles, triangles and half circles. Most floor plans, yards and walls are composite shapes. There is no single formula for them, but you can **split** them into shapes you already know.

Split and add

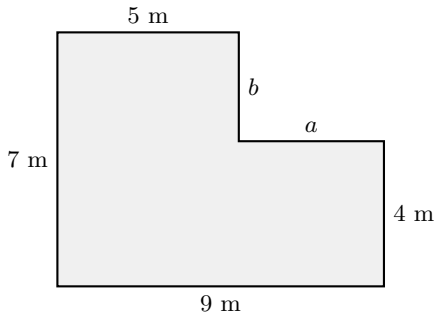
Step 1. Draw lines to split the shape into rectangles, triangles or circle parts.
Step 2. Find any missing side lengths.
Step 3. Find the area of each part.
Step 4. Add the areas. Write the answer in square units (m^2, ft^2, ...).

A **semicircle** is half a circle: $A = \frac{1}{2}\pi r^2$ (the radius r is half the diameter).

Watch out. The parts must **not overlap**. Each square unit is counted only once. To find a missing side, look across the shape: the short sides on one side add up to the long side on the other side.

Example 1

The floor plan shows the seating area of a coffee shop. All the corners are square corners (90°).

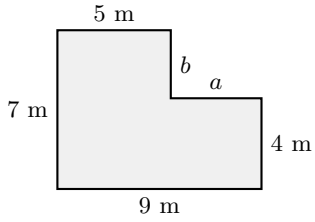


(a) Find the missing side lengths a and b.

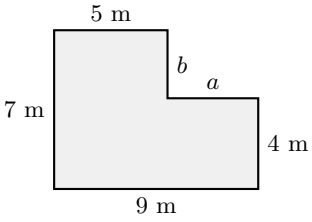
$a =$ _____ $b =$ _____

Hint. The top (5 m) and a together are as long as the bottom (9 m).

(b) Split the floor with a **vertical** (up-and-down) line into two rectangles. Find the area of the floor.



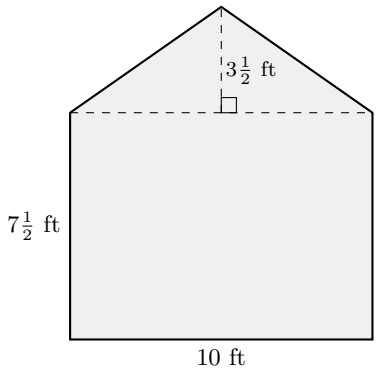
(c) Split it another way, with a **horizontal** (side-to-side) line. Do you get the same area?



Fractions of a foot. Change a fraction to a decimal before you use the calculator: $\frac{1}{4} = 0.25$, $\frac{1}{2} = 0.5$, $\frac{3}{4} = 0.75$. So $7\frac{1}{2}$ ft $= 7.5$ ft.

Example 2

Mila is putting new siding on the end wall of her garden shed. The wall is a rectangle with a triangle on top.



(a) Find the area of the rectangle and of the triangle.

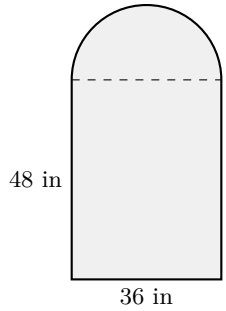
(b) What is the total area of the wall?

(c) The shed has two end walls like this. One box of siding covers 50 ft^2. How many boxes does Mila need for both end walls?

Calculator. For the semicircle in Example 3: 0.5 [×] [π] [×] 18 [x^2] [ENTER] shows 508.938... Keep all the digits until the last step, then round.

Example 3

An arched window is a rectangle 36 in wide and 48 in tall, with a semicircle on top.



(a) Find the area of the rectangle.

(b) Find the area of the semicircle.

(c) Find the total area of the window, to the nearest square inch.

(d) Window film is sold by the square foot. Change your answer to square feet, to the nearest tenth. (1 ft^2 = 144 in^2)

2. Whole Minus the Cut-Out

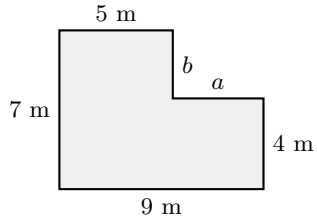
Some shapes have a hole or a piece missing: a lawn with a pool in it, a wall with a door and a window. For these, find the area of the **whole** shape, then **subtract** the area of each **cut-out** (the part that is not included).

Area = area of the whole shape − area of the cut-out(s)

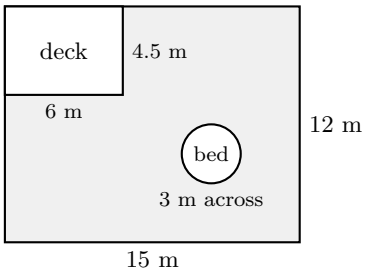
Many composite shapes can be done either way: split and add, *or* whole minus cut-out. Using both is a good way to check.

Example 4

(a) Find the area of the coffee shop floor from Example 1 using whole minus cut-out.



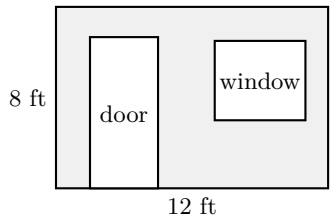
(b) The Nguyens' backyard is a rectangle 15 m by 12 m. It has a deck 6 m by 4.5 m and a round flower bed 3 m across. The rest will be sod (grass). Find the area of sod, to the nearest hundredth of a square metre.



(c) Sod costs \$4.80 per m^2. The landscaper rounds the area **up** to the next whole square metre. What will the sod cost (before tax)?

Example 5

Owen is painting one wall of his bedroom. The wall is 12 ft long and 8 ft high. It has a door 3 ft wide and 6 ft 8 in high, and a window 4 ft wide and 3 ft 6 in high. He will not paint the door or the window.



(a) Change the door height and the window height to feet.

6 ft 8 in = _____ 3 ft 6 in = _____

(b) Find the area Owen will paint.

(c) Owen will paint two coats. One can of paint covers about 100 ft^2. How many cans does he need?

3. Irregular Shapes: Counting Squares

An **irregular shape** has curved or uneven sides, like a pond, a lake or a spill. There is no formula for it, so we **estimate** its area with a grid.

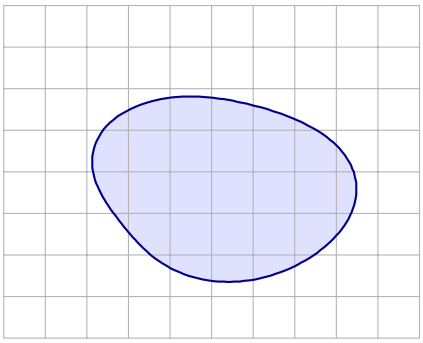
- A **full square** is completely inside the shape.
- A **part square** is cut by the edge of the shape. Some part squares are almost full and some are almost empty, so on average a part square is about **half** full. Two part squares make about one full square.

Area $\approx$ (full squares $+$ part squares $\div 2$) $\times$ area of one square

Count carefully. Put a small mark in each square as you count it (a tick for full, a dot for part), so you do not count a square twice or miss one.

Example 6

A park map shows a pond on a grid. Each square is 1 m by 1 m, so each square is 1 m^2.



(a) Complete the table.

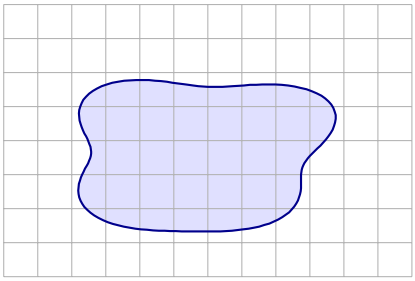
Full squares	
Part squares	
Part squares ÷ 2	
Estimate of area	

(b) Finish the sentence. The pond's area is about _____.

When each square stands for more than 1 square unit, find the area of **one square** first, then multiply. To check an estimate, draw a rectangle **around** the shape (the real area is less) and a rectangle **inside** it (the real area is more). A good estimate is between the two.

Example 7

A farmer's map shows a slough (a prairie pond) in a field. On the map, each square stands for 10 m by 10 m.



(a) What area does one square stand for?

(b) Estimate the area of the slough.

(c) On the map, draw the smallest rectangle (on grid lines) that goes all the way **around** the slough. Find the area it stands for.

(d) Draw the biggest rectangle that fits **inside** the slough. Find its area. Is your estimate from (b) between the two rectangles?

4. Is My Answer Reasonable? Checking a Second Way

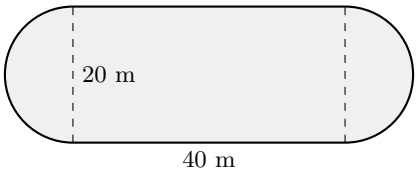
Area mistakes are easy to make: a missing side, an overlap, a diameter used as a radius. Always check your answer.

Ways to check an area

- Do it a **second way**: split the shape a different way, or use whole minus cut-out. You should get the same answer.
- Compare with a **rectangle around** the shape: the answer must be **less** than the rectangle's area.
- Check the **units**: area is always in square units (m^2, ft^2, ...).

Example 8

A town floods an outdoor skating oval. It is a rectangle 40 m long and 20 m wide, with a semicircle on each end.



(a) Estimate first. What are the length and width of a rectangle that goes around the whole oval? What is its area?

(b) Find the area of the oval, to the nearest square metre. Is it less than your answer to (a)?

(c) Jordan got 2057 m^2. Use part (a) to explain how you know his answer is wrong. What mistake do you think he made?

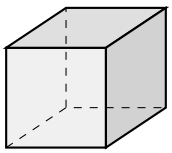
Lesson 10: Nets and Surface Area of Prisms

Unfolding boxes and other prisms into flat nets, and adding the areas of all the faces to find how much material covers a box, a tent or a room.

1. Prisms, Faces and Nets

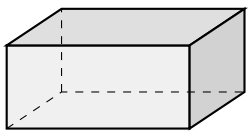
A **3-D object** (three-dimensional object) has length, width and height, like a box. Each flat surface of a 3-D object is a **face**. Two faces meet along an **edge**. Edges meet at a corner called a **vertex** (plural: **vertices**).

A **prism** is a 3-D object with two matching ends that face each other. The ends are called the **bases**. All the other faces are rectangles. A prism is named for the shape of its bases. In the pictures, **dashed lines** are edges hidden at the back.



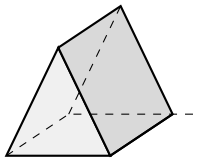
cube

6 equal squares



rectangular prism

bases are rectangles



triangular prism

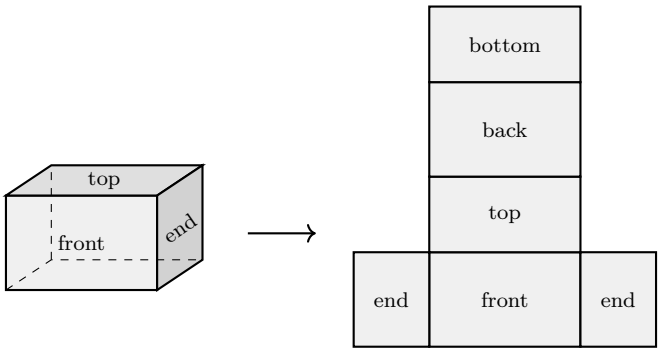
bases are triangles

Example 1

Complete the table for the three prisms above.

	Cube	**Rectangular prism**	**Triangular prism**
Number of faces			
Shapes of the faces			
Number of edges			
Number of vertices			

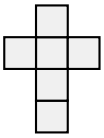
A **net** is a flat pattern that folds up to make a 3-D object. Each face of the object appears in the net **once**, with no gaps and no overlaps. Think of a cereal box cut open along some edges and pressed flat.



Example 2

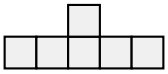
Each pattern is made of 6 squares. Which patterns are nets of a cube? Imagine folding each one. Write yes or no.

(a)



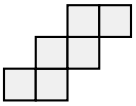
_____

(b)



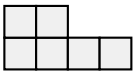
_____

(c)



_____

(d)



_____

2. Surface Area of a Rectangular Prism

The **surface area** (SA) of a 3-D object is the total area of all of its faces. It is the same as the area of its net. Surface area tells you how much material covers the outside: cardboard for a box, wrapping paper, paint or fabric. It is measured in square units, such as cm^2, m^2, in^2 or ft^2.

A rectangular prism has 6 faces in 3 matching pairs: top and bottom, front and back, and the two ends. A **face table** keeps track of them so that no face is missed or counted twice.

Surface area = the sum of the areas of all the faces = the area of the net.

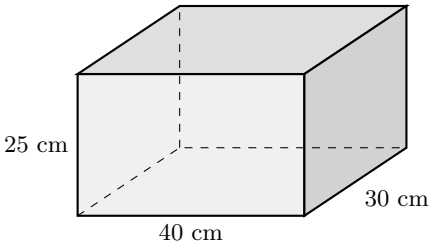
Rectangular prism (length l, width w, height h): $SA = \underbrace{2lw}_{\text{top \& bottom}} + \underbrace{2lh}_{\text{front \& back}} + \underbrace{2wh}_{\text{two ends}}$

Cube (edge s): $SA = 6s^2$ (6 equal squares, each $s \times s$)

Watch out. Surface area is *not* $l \times w \times h$. Multiplying all three lengths gives the **volume** (how much fits inside, in Unit 4). Surface area *adds* the areas of the faces.

Example 3

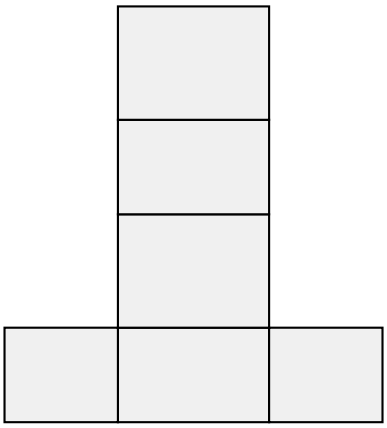
A shipping box is 40 cm long, 30 cm wide and 25 cm high.



(a) Write the size of each face on the net.

(b) Complete the face table.

Faces	**Size**	**Area of one**	**Area of two**
top & bottom			
front & back			
two ends			
		Surface area	



(c) Check your answer with the formula.

(d) Write the surface area in m^2.

Calculator. For a cube, use the $\boxed{x^2}$ key: an edge of 2.5 in gives $6 \boxed{\times} 2.5 \boxed{x^2} \boxed{\text{ENTER}} = 37.5$.

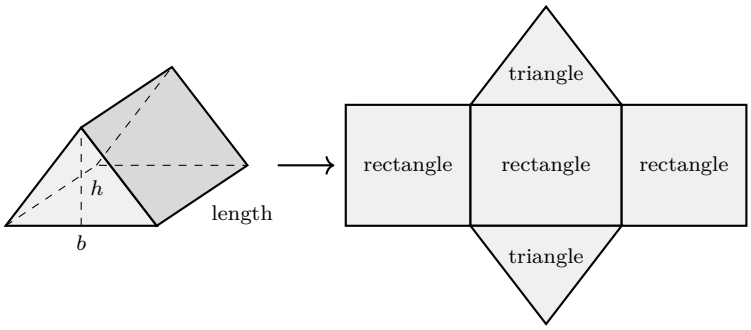
Example 4

(a) A wooden toy block is a cube with edges $2\frac{1}{2}$ in long. Find its surface area.

(b) A storage ottoman is a cube with edges 1 ft 3 in long. Fabric covers all 6 faces. How much fabric is needed, in in^2 and in ft^2?

3. Surface Area of a Triangular Prism

A triangular prism has **5 faces**: 2 matching triangles (the bases) and 3 rectangles. Each rectangle is as long as the prism. Its width is one side of the triangle. So the net is three rectangles in a row with a triangle on each side.



Triangular prism: $SA = 2$ triangles $+ 3$ rectangles

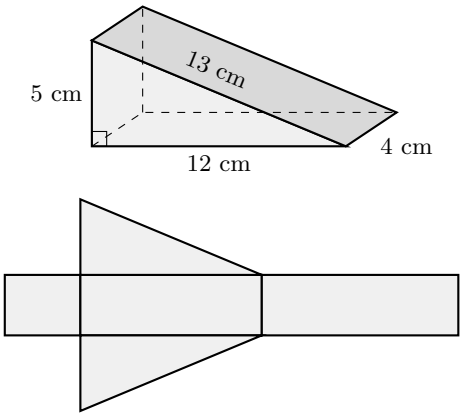
Each triangle: $A = \frac{1}{2}bh$, using the **height** of the triangle (straight up from the base).
Each rectangle: $A =$ (side of the triangle) $\times$ (length of the prism).

Shortcut: the 3 rectangles together = (perimeter of the triangle) $\times$ (length of the prism).

Watch out. The height of the triangle is *not* one of its slanted sides (unless the triangle has a right angle). Use the height for the two triangles, and the three sides for the three rectangles.

Example 5

A rubber door wedge is a triangular prism. Its ends are right triangles with sides 5 cm, 12 cm and 13 cm. The wedge is 4 cm wide.

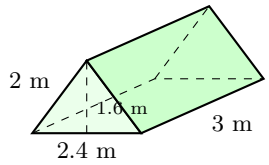


(a) On the net, write how to find the area of each face.

(b) Find the surface area of the wedge.

Example 6

Jordan is sewing a small camping tent shaped like a triangular prism. The front is a triangle 2.4 m wide at the bottom and 1.6 m high. Each sloping side of the tent is 2 m. The tent is 3 m long.



(a) How much fabric is needed for the whole tent, including the floor?

(b) The floor is made of a heavier fabric. How much of the lighter nylon is needed for the rest?

(c) The nylon costs \$8.95 per m^2. What does the nylon cost with 5% GST?

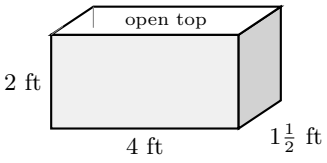
4. Open Boxes and Real Jobs

Not every job covers every face. An **open box** has no top, so it has only 5 faces. When you paint a room you paint the walls (and maybe the ceiling), but not the floor, the doors or the windows. Always ask first: *which faces get covered?*

Covering a real object
Step 1. Sketch the object. List only the faces that get covered.
Step 2. Find the area of each of those faces and add.
Step 3. Subtract any openings that are not covered (doors, windows).
Step 4. Multiply by the number of coats (for paint or stain).
Step 5. Divide by the area that one can, roll or sheet covers. Round **up**.
Step 6. Find the cost.

Example 7

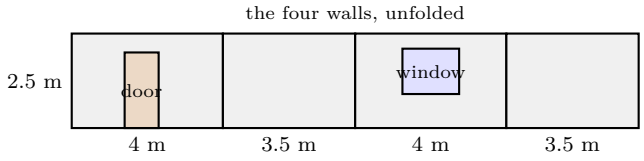
Amira is building a planter box with no top out of plywood. The box is 4 ft long, $1\frac{1}{2}$ ft wide and 2 ft high.
(a) Which faces does she need to build? Find the area of plywood needed.



(b) A sheet of plywood is 4 ft by 8 ft. Is one sheet enough? Explain, using area.

Example 8

Lucas is painting the four walls of his bedroom. The room is 4 m long, 3.5 m wide and 2.5 m high. He does not paint the ceiling, the door (0.9 m by 2 m) or the window (1.5 m by 1.2 m). He wants 2 coats. A can of paint covers about 35 m^2 and costs \$54.98.



(a) Find the area of the four walls.

(b) Subtract the door and the window.

(c) How many cans does he need for 2 coats?

(d) Find the cost of the paint with 5% GST.

Lesson 11: Surface Area of Cylinders, Pyramids and Cones

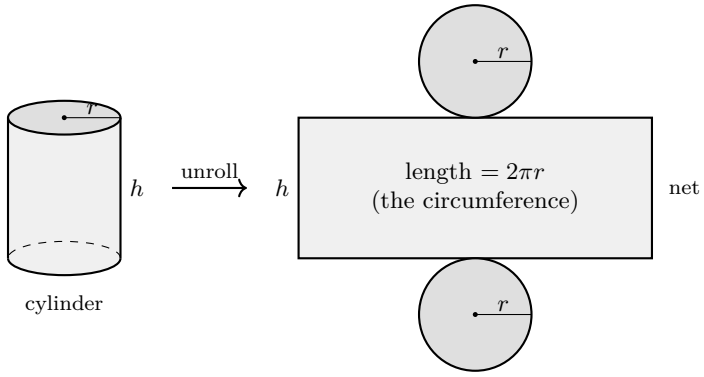
Unrolling a cylinder, unfolding a pyramid and flattening a cone, then adding the areas of the pieces to find how much material covers an object.

1. The Net of a Cylinder

A **cylinder** has two circles of the same size (the top and the bottom) and one curved side. Think of a can of soup. Peel off the label and lay it flat: it is a **rectangle**. So the **net** of a cylinder is one rectangle and two circles.

- The rectangle's width is the **height** of the cylinder, h.
- The rectangle's length wraps once around the can, so it is the **circumference** of the circle: $C = 2\pi r$ (or πd).

The **surface area** (SA) of a solid is the total area of all the pieces of its net. It is in square units.



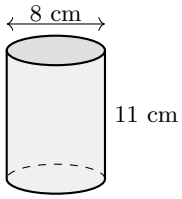
Each circle: $A = \pi r^2$ **Curved side** (the label): $A = \text{length} \times \text{width} = 2\pi r \times h = 2\pi rh$
Closed cylinder: $\text{SA} = 2\pi r^2 + 2\pi rh$ (two circles + the side)

Calculator. Find each part, write it down, then add. Keep all the decimals until the end. For $r = 4$ and $h = 11$: 2 [×] [π] [×] 4 [x^2] [ENTER] gives 100.53...; 2 [×] [π] [×] 4 [×] 11 [ENTER] gives 276.46...

Watch out. If you are given the **diameter**, halve it to get the radius. And $2\pi r^2$ means $2 \times \pi \times r \times r$: only the r is squared.

Example 1

A can of soup is 8 cm across (its diameter) and 11 cm tall.
(a) Fill in the measurements of the net.
The rectangle is _____ cm long and _____ cm wide.
Each circle has a radius of _____ cm.



(b) Find the area of the label (the curved side), to the nearest tenth.

(c) Find the surface area of the whole can, to the nearest tenth.

Example 2

Rhys is painting the outside of an old steel drum (with both ends) to use as a rain barrel. The drum has a diameter of $22\frac{1}{2}$ in and a height of $34\frac{1}{2}$ in.

(a) Find the surface area in square inches, to the nearest whole number.

(b) Change the area to square feet, to the nearest tenth. (1 $\text{ft}^2 = 144\ \text{in}^2$)

(c) One can of spray paint covers about 20 ft^2. How many cans does Rhys need for one coat?

2. Open Cylinders: Use Only the Parts You Need

Not every cylinder has two ends. A pipe has no ends. A pail has a bottom but no top. Before you calculate, ask: **which pieces are really there**, or which pieces really get covered?

Closed (a can with a lid, a drum)	2 circles + side	$SA = 2\pi r^2 + 2\pi rh$
Open at one end (a pail, a planter, a stock tank)	1 circle + side	$SA = \pi r^2 + 2\pi rh$
No ends (a pipe, a label, the walls of a grain bin)	side only	$SA = 2\pi rh$

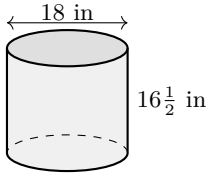
Example 3

(a) Which pieces of the net does each object need? Write "2 circles + side", "1 circle + side" or "side only".

a tube of tennis balls with a lid _____ a metal drainpipe _____

a round cake pan _____ the paper label on a jar _____

(b) Amara is covering a round storage ottoman with fabric. The fabric covers the top and the side, but not the bottom. The ottoman has a diameter of 18 in and a height of $16\frac{1}{2}$ in. How much fabric covers it, in in^2 (to the nearest whole number) and in ft^2 (to the nearest tenth)?



Example 4

A parkade in Calgary has 12 round concrete columns. Each column has a diameter of 0.6 m and a height of 3.2 m. Only the curved sides get painted, with 2 coats. One can of paint covers 37 m^2 and costs $68.99.

(a) Find the area of the side of one column.

(b) Find the total area to paint (12 columns, 2 coats).

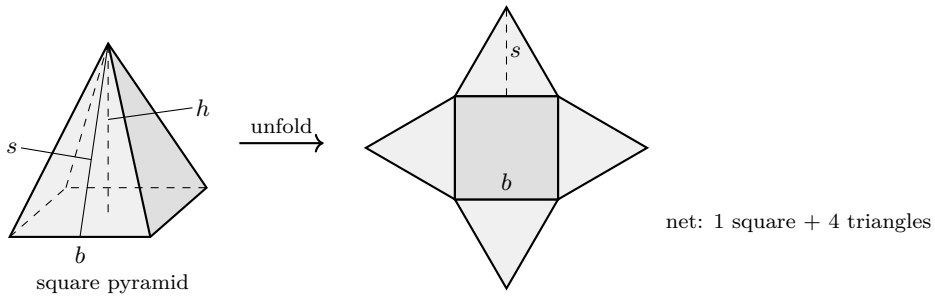
(c) How many cans of paint are needed?

(d) What do the cans cost with 5% GST?

3. Square Pyramids

A **square pyramid** has a square base and four triangles that meet at a point at the top, called the **apex**. In this booklet the four triangles are always the same size. Unfold it and the net is a square with a triangle on each side.

The **slant height** s is the height of one triangle face. It goes from the middle of a base edge, up the face, to the apex. The **height** h of the pyramid goes straight up the *inside*, from the centre of the base to the apex. Surface area uses the slant height, not the height.

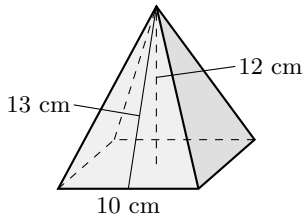


Base: $b \times b = b^2$	**One triangle:** $\frac{1}{2} \times b \times s$
Square pyramid: $\text{SA} = b^2 + 4 \times \frac{1}{2}bs$ (base + 4 triangles; this is the same as $b^2 + 2bs$)	
A roof, or a tent with no floor, has **no base**: use only the 4 triangles.	

Watch out. Use the **slant height** s, not the height h. The slant height is on the outside of the face, and it is always longer than the height. If a picture shows both, the height is the dashed line inside.

Example 5

(a) Find the surface area of the square pyramid. Which length is *not* used?

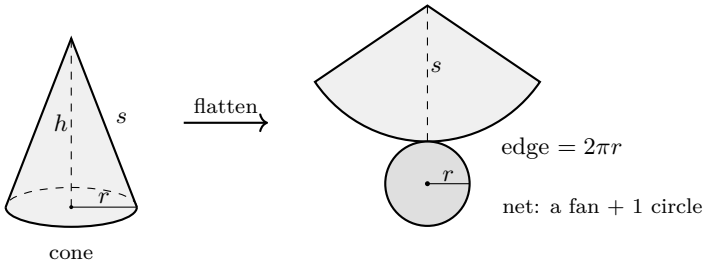


(b) Jonah's camping tent is a square pyramid with a sewn-in floor. The floor is 7 ft by 7 ft, and the slant height of each wall is $6\frac{1}{2}$ ft. How much fabric is in the tent (4 walls and the floor)?

4. Cones

A **cone** has a circle for a base and a curved side that comes to a point, the **apex**. The **slant height** s goes from the apex, down the outside, to the edge of the base.

Cut the curved side along a slant line and flatten it: you get a **fan** shape. Think of the fan as many thin triangles. Each one has a height of s, and all their bases together go around the circle, $2\pi r$. So the fan's area is $\frac{1}{2} \times 2\pi r \times s = \pi rs$.



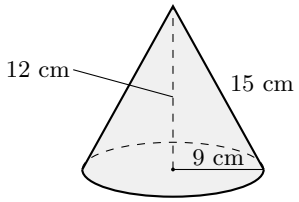
Base: πr^2	**Curved side:** πrs
Closed cone: $\text{SA} = \pi r^2 + \pi rs$	
An **open** cone (an ice-cream cone, a funnel, a party hat, a roof) has no base: use only πrs.	

Watch out. Just like the pyramid, use the slant height s, not the height h. And use the radius: if you are given the diameter, halve it first.

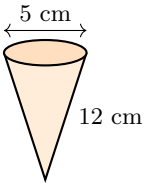
Example 6

Round each answer to the nearest tenth.

(a) Find the surface area of the closed cone.



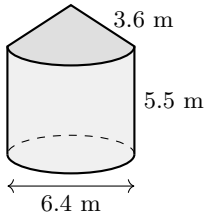
(b) A waffle cone for ice cream is open at the top. It is 5 cm across the top and has a slant height of 12 cm. How much waffle is in it?



Some objects are made of two solids, like a grain bin with a cone roof. Find the area of each part you need, then add. Leave out the faces that are hidden or not there, like the floor of a bin that sits on concrete.

Example 7

A farmer near Three Hills is building a steel grain bin. The walls are a cylinder with a diameter of 6.4 m and a height of 5.5 m. The roof is a cone with a slant height of 3.6 m. The floor is concrete, so it needs no steel.



(a) Find the area of the walls.

(b) Find the area of the roof.

(c) Find the total area of steel, to the nearest tenth.

(d) The steel costs \$24.80 per m^2. Use your answer to (c) to find the cost before tax.

Lesson 12: Unit 3 Review

Every rule from the unit on one page, worked examples that mix the topics, and a practice unit test.

1. The Unit on One Page

Use this page to study. Each row is one skill: what to do, and a short example. Conversion numbers marked ≈ are rounded, so answers that use them are close, not exact.

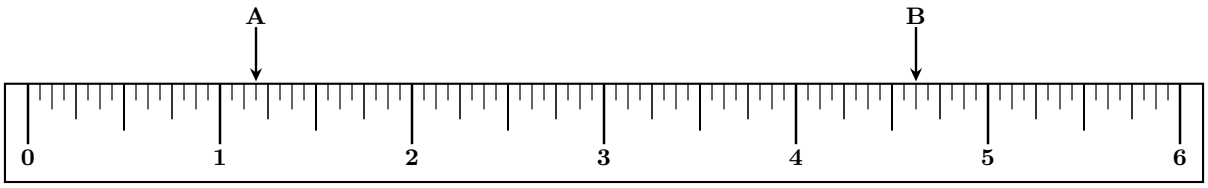
Skill	What to do	Example
Mixed numbers	Add or subtract: get a common denominator, then do the whole numbers and the fractions. Borrow 1 as $\frac{8}{8}$ (or $\frac{4}{4}$, ...) when you need to. Multiply: change to improper fractions first. Simplify.	$2\frac{3}{4} + 1\frac{5}{8} = 2\frac{6}{8} + 1\frac{5}{8} = 3\frac{11}{8} = 4\frac{3}{8}$
SI length	km, hm, dam, **m**, dm, cm, mm. Each step down: $\times 10$. Each step up: $\div 10$. 1 km = 1000 m; 1 m = 100 cm = 1000 mm.	3.45 m = 345 cm (2 steps down) 850 m = 0.85 km (3 steps up)
Imperial length	12 in = 1 ft; 3 ft = 1 yd; 36 in = 1 yd; 5280 ft = 1 mi; 1760 yd = 1 mi. To a smaller unit, multiply. To a bigger unit, divide.	100 in: $100 \div 12 = 8$ R 4, so 8 ft 4 in
Feet and inches	$5'\,9''$ means 5 ft 9 in. The inches part is less than 12. Adding: change 12 in into 1 ft. Subtracting: borrow 1 ft as 12 in.	$5'\,9'' + 2'\,8'' = 7'\,17'' = 8'\,5''$
Reading a ruler	Whole inches: the last number you passed. Then count the $\frac{1}{16}$ spaces past it and simplify.	12 spaces past 3: $3\frac{12}{16} = 3\frac{3}{4}$ in
SI ↔ imperial	1 in = 2.54 cm (exact); 1 ft = 30.48 cm; 1 yd = 0.9144 m; 1 m ≈ 3.28 ft; 1 mi ≈ 1.609 km; 1 km ≈ 0.621 mi. The smaller unit always has the bigger number.	$5'\,10'' = 70$ in $70 \times 2.54 = 177.8$ cm
Perimeter and circumference	P = the sum of the sides; rectangle $P = 2l + 2w$. Circle: $C = \pi d = 2\pi r$. Curved edge of a half circle = $\pi d \div 2$. Pieces sold in set lengths: round **up**.	$d = 30$ cm: $C = \pi \times 30 \approx 94.2$ cm
Rearranging formulas	Undo + or − first, then × or ÷. $w = (P - 2l) \div 2$; $d = C \div \pi$; $w = A \div l$. Check by putting the answer back in.	$P = 36$, $l = 11$: $w = (36 - 22) \div 2 = 7$
Area	Rectangle $A = lw$; parallelogram $A = bh$; triangle $A = \frac{1}{2}bh$; trapezoid $A = \frac{1}{2}(a + b)h$; circle $A = \pi r^2$. Answers in square units.	$r = 5$ m: $A = \pi \times 5^2 \approx 78.5$ m^2
Area units	1 ft^2 = 144 in^2; 1 yd^2 = 9 ft^2; 1 m^2 = 10 000 cm^2; 1 m^2 ≈ 10.76 ft^2.	54 ft^2 $\div 9 = 6$ yd^2
Composite and irregular shapes	Split into simple shapes and add, or take the whole shape minus the cut-out. Irregular: (full squares + part squares $\div 2$) × area of one square.	12 full, 20 part: $12 + 20 \div 2 = 22$ squares
Surface area of prisms	Draw the net. Find the area of every face, then add. A box has 3 pairs of equal faces. An open box has no top.	Cube with 3 cm edges: $6 \times (3 \times 3) = 54$ cm^2
Surface area of cylinders, pyramids, cones	Cylinder: $SA = 2\pi r^2 + 2\pi rh$ (the side unrolls to a rectangle). Square pyramid: base + $4 \times \frac{1}{2}bs$. Cone: $SA = \pi r^2 + \pi rs$. s is the **slant height**.	Cylinder, $r = 2$ cm, $h = 5$ cm: $2\pi(2^2) + 2\pi(2)(5) \approx 88.0$ cm^2

Remember. Put every length in the **same unit** before you add or multiply. Use the radius in πr^2, not the diameter. Perimeter is a length (m, ft); area and surface area are in **square** units (m^2, ft^2). Round only at the **end**. Round **up** when you buy pieces, cans or rolls; round **down** when you count the pieces you can cut.

2. Measuring and Converting Length

Example 1

The ruler is real size.



(a) Read the length at each arrow.
A _____ B _____

(b) How far apart are the two arrows?

(c) Sofia needs 3 pieces of baseboard, each $2\frac{3}{4}$ ft long. What total length is that? Is one 8 ft piece enough?

(d) She buys a 10 ft piece instead. How much is left after she cuts the 3 pieces? (Ignore the saw cuts.)

Example 2

(a) Fill in the blanks.

3.2 km = _____ m
65 mm = _____ cm
0.48 m = _____ cm

(b) Fill in the blanks.

9 ft 5 in = _____ in
2 mi = _____ yd
7 yd = _____ ft

(c) Liam is $5' 11''$ tall. What is his height in centimetres, to the nearest centimetre?

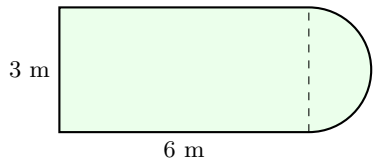
(d) Ruby runs a 5 km fun run. Estimate the distance in miles, then find it to the nearest tenth. (1 km $\approx$ 0.621 mi)

(e) Two shelves are $3' 9''$ and $2' 10''$ long. Will they fit side by side on a wall space that is $6' 6''$ wide?

3. Perimeter, Circumference and Rearranging

Example 3

A garden bed is a rectangle with a half circle on one end.



(a) Find the perimeter of the bed, to the nearest hundredth of a metre.

(b) Edging is sold in 2.4 m pieces for $7.98 each. How many pieces are needed, and what do they cost?

(c) Find the total with 5% GST.

Example 4

(a) A rectangular dog run uses 32 m of fencing. It is 9.5 m long. How wide is it? Check your answer.

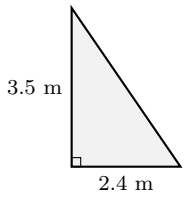
(b) The edge of a round table top is 4.4 m around. Find its diameter, to the nearest hundredth of a metre.

(c) A rectangular patio has an area of 22.5 m^2. It is 3 m wide. How long is it?

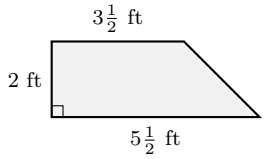
4. Area

Example 5

(a) A triangle-shaped sail has a base of 2.4 m and a height of 3.5 m. Find its area.



(b) A flower bed is a trapezoid. Its parallel sides are $3\frac{1}{2}$ ft and $5\frac{1}{2}$ ft, and they are 2 ft apart. Find its area.



(c) A round rug is 1.8 m across. Find its area, to the nearest hundredth.

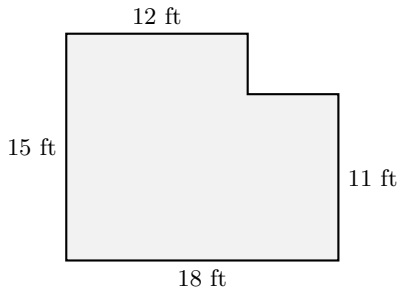
(d) Fill in the blanks.

9 ft^2 = _____ in^2 45 ft^2 = _____ yd^2

4.2 m^2 = _____ cm^2

Example 6

Jaspreet is putting carpet in an L-shaped family room. All the corners are square.



(a) Find the area as the **whole rectangle minus the cut-out** corner.

(b) Check by splitting the room into two rectangles.

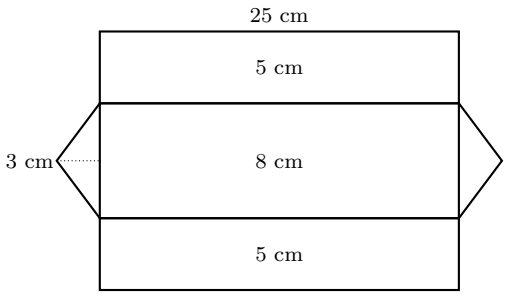
(c) Carpet costs $38.50 per square yard and is sold in whole square yards. What will the carpet cost before tax?

5. Surface Area

The **surface area** (SA) of a 3-D object is the total area of all its faces. Unfold the object into a flat **net**, find the area of each part, and add.

Example 7

This is the net of a chocolate box shaped like a triangular prism. The triangle ends have sides of 5 cm, 5 cm and 8 cm and a height of 3 cm. The box is 25 cm long.

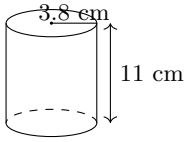


(a) Complete the face table.

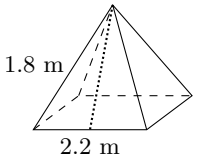
Faces	**Area**
2 triangles	
bottom (8 cm by 25 cm)	
2 sides (5 cm by 25 cm)	
Surface area	

(b) The factory has a sheet of cardboard with an area of 1 m^2 (10 000 cm^2). If no cardboard is wasted, how many whole boxes can it make?

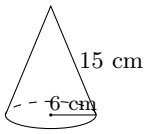
Example 8



(a), (b) a soup can



(c) a tent



(d) a funnel

(a) A soup can has a radius of 3.8 cm and a height of 11 cm. Find the area of the paper label that wraps around the side.

(b) Find the surface area of the whole can (side, top and bottom).

(c) A tent is a square pyramid with a 2.2 m by 2.2 m floor. The slant height of each side is 1.8 m. How much fabric is in the 4 sides and the floor?

(d) A kitchen funnel is a cone with a radius of 6 cm and a slant height of 15 cm (ignore the spout). It has no base. Find the area of plastic in the cone.